European Commission

Joint Research Centre

Blue Mirror

One Water - One Health Monthly Chronicle



OCTOBER 2025
VOL 1 ISSUE 2

GLOWACON news:
Aviation Programme
p. 16

Monthly Bulletin
p. 23

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Contact information
Name: JRC Water Laboratory
Address: Joint Research Centre, Via E. Fermi, Ispra, Varese (VA)
Email: JRC-EU4S-DEEP@ec.europa.eu
Website: https://wastewater-observatory.jrc.ec.europa.eu/

The Joint Research Centre: EU Science Hub
https://joint-research-centre.ec.europa.eu

JRC144490

PDF ISSN 3094-5836 KJ-01-25-569-EN-N

Luxembourg: Publications Office of the European Union, 2025

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How to cite this report: Gawlik, B.M., Panzarella, G., Comero, S., Tessarolo, A., Koutsolioutsou, A. et al., *Blue Mirror - One Water - One Health Monthly Chronicle - October 2025 - Vol. 1 Issue 2*, Publications Office of the European Union, Luxembourg, 2025, JRC144490.

EDITORIAL OFFICE
Joint Research Centre, Via E. Fermi, Ispra, Varese (VA)
email: JRC-EU4S-DEEP@ec.europa.eu
https://wastewater-observatory.jrc.ec.europa.eu/

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CHIEF EDITOR
Bernd Manfred Gawlik

ASSOCIATE EDITOR
Giulia Panzarella

SECTION EDITOR
Sara Comero

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EDITORIAL
1 *Between Sirens and Stars*: the Song of Science and Democracy

CORRESPONDANCE
3 Flushing Out Myths: Rethinking Equity in Wastewater Surveillance from an LMIC Lens

COMMENT
7 *Human Habitats*: Designing for a Changing Planet

10 The Risk Management Game: Building Water Reuse Governance Through Simulation and Play

12 Destruction, Technology, Creativity

EU SUPER-SITES: INSIGHTS FROM THE OUTSIGHT
14 Madeira Giga-Site

GLOWACON NEWS
16 Aviation Wastewater Surveillance Programme

THE HUMAN SAMPLE
17 Wastewater and Environmental Surveillance under One Water–One Health Approach

BRIEF COMMUNICATIONS
20 News

EU WASTEWATER OBSERVATORY FOR PUBLIC HEALTH
23 **MONTHLY BULLETIN**

EDITORIAL

Between Sirens and Stars: The Song of Science and Democracy

Bernd Manfred Gawlik

On our cover, a siren rises from the depths: half woman, half song, half question. Since the dawn of myth, Sirens have haunted the human imagination: symbols of beauty and peril, of knowledge that tempts and truths that undo.

For Ulysses, the Sirens were the embodiment of desire: voices so exquisite they promised revelation yet offered only shipwreck. He chose to listen but not to yield, binding himself to the mast as his crew rowed past the fatal coast. His act was both defiance and surrender: the yearning to know, restrained by the will to survive.

Pythagoras, by contrast, heard other Sirens, not of the sea, but of the spheres. His Sirens sang in numbers and intervals, their harmonies echoing through the motion of planets and stars. Where Ulysses' Sirens lured sailors toward the rocks, Pythagoras' Sirens lifted minds toward the heavens. One song destroys; the other orders. Both reveal the perilous power of listening.

Between these two myths stretches the human condition: we are beings drawn to sound, to meaning, to the invisible vibrations that promise understanding. Whether we seek knowledge in the song of the sea or in the harmony of the cosmos, we are still tied to the mast, still straining to hear what lies beyond the *noise of our age*.

Perhaps this is why the siren remains an enduring emblem: she reminds us that curiosity and danger share the same melody, and that the pursuit of truth, scientific, artistic, or spiritual, is always sung in a key that can both save and undo us.

The Sirens of Science

Like the sirens of myth, science calls to us with the promise of knowledge and understanding. But unlike those dangerous voices that lured sailors to illusion, the song of science is grounded in doubt, testing, and correction. It reminds us that **science is credible not because it claims perfection, but because it constantly challenges itself**. It helps us grasp reality, act effectively, and make accurate predictions, yet it always remains open to revision when new evidence arises. Its strength lies not in the purity of individual intentions, but in collective scrutiny: in peer review, open debate, and the replicability of results. We do not believe in science as an act of faith; we trust it as a self-correcting system - one that learns from its errors and refines its truth, note by note, as the song continues.

This is made possible by the institutions devoted to that very mission: universities and research centres like ours, as well as scientific journals and peer-review processes. When no consensus yet exists within the scientific community, exploring diverse approaches and testing alternative hypotheses is crucial. But when consensus is broad, such as on the human-driven causes of climate change, as highlighted in this issue's article "*Human Habitats: Designing for a Changing Planet*", giving attention to discredited views is not a mark of intellectual openness; it amounts to spreading misinformation. Climate denialism is not new. Such theories have been refuted and archived, as have many others before them.

EDITORIAL
Between Sirens and Stars: The Song of Science and Democracy

Bernd Manfred Gawlik

As Senator Elena Cattaneo reminds us: "How many times might greater scientific competence have helped politics legislate better, protecting citizens and public resources? Xylella, Stamina, GMOs, vaccines, animal testing - these are only a few examples [1]."

Navigating Knowledge

We must work to ensure that scientific awareness aligns with political intention.

At the foundation lies what we might call *epistemic civic education*: the ability to navigate an informational ecosystem that is noisy, crowded, and often polluted.

"The modern world overwhelms people with an excess of information that is disorienting and dangerous to the mind," wrote the Swiss bibliographer Conrad Gessner in 1550, referring, to the newly invented printing press. *Blue Mirror* exists in this same spirit: to bring clarity, to unite the scientific community around essential themes, e..g., the preservation of ecosystems, preparedness for future "Disease X" events, the value of wastewater in early pathogen detection and pandemic prevention, water reuse, and circular economy models.



Figure A
Source: Canva

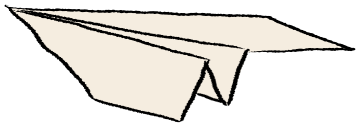
Figure A| "When the sea sings, it is the Siren; when the sea speaks, it is Triton". Son of Poseidon and Amphitrite, Triton used to blow into seashells to communicate with the sea itself. From this comes the evocative idea that "listening to a seashell" means listening to Triton: the voice of the sea.

Scientific knowledge saves us, forming the very foundation of a functioning democracy.

This chronicle aspires to be a scientific-democratic space for communicating *One Health* science, with a particular focus on the waters of our planet - waters that are both mirror and message, source and archive of life itself. In these depths, we might recall *Tritone*, the ancient *messenger of the sea*, whose conch carried the voices of the ocean to the world above. To navigate these waters today is to follow his example: to listen, with attention and humility, to the many sounds of the Earth's blue body. Learning to hear this resonant call, to translate its evidence into action, is not only an act of science. It is an act of democracy.

CORRESPONDANCE

Flushing Out Myths: Rethinking Equity in Wastewater Surveillance from an LMIC Lens



Fatma Z. Guerfali, Assistant Professor at Institut Pasteur de Tunis

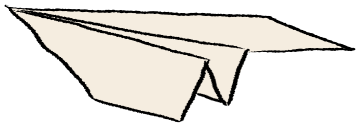
Wastewater doesn't lie, but sometimes, the stories we tell about it do. Since the COVID-19 pandemic hit our health systems, wastewater surveillance has become a darling of global health surveillance and preparedness, celebrated as a crystal ball for early warning, and a marvel of cross-sector collaboration. From polio to SARS-CoV-2 or antimicrobial resistance (AMR) surveillance, the idea that sewage can warn us before sickness spreads is both brilliant and strangely attractive. But let's face it: as enthusiasm flushes through the global system, so do assumptions about what equity means, when taken from a low- and middle-income countries (LMICs) perspective. As someone working in and with LMIC contexts I've come to see words like capacity gap, infrastructure deficit, and lack of readiness sticking as uncomfortable truths in global conversations. Of course, they are not all being a pure fiction, nor do they come from malicious claims. Yet, like blockages in a pipeline, they prevent real flow, and equitable access to decision-making. So perhaps it's time to unclog the narrative and take a step back in this short reflection to explore some of the myths and realities of what equity in wastewater surveillance really means from an LMIC lens, where innovation often thrives despite scarcity, and where equity isn't just a buzzword, but a daily negotiation.

MYTH 1: "Equity means giving everyone the same tools".

Not quite. Handing out the same kits, protocols, or dashboards to all countries assumes identical contexts. In reality, equity should mean tailoring support, not cloning systems. The analytical tools that work perfectly in a high-income setting might crumble in a LMIC if the electricity goes out every afternoon. In LMICs, true equity means contextualizing the reality,
not copy-pasting a high-income countries system. From this perspective, equity isn't about matching budgets. It's about matching solutions with contexts. After all, across LMICs, scientists, engineers, and public health professionals have often been practicing what I am tempted to call frugal surveillance: creative, low-cost approaches, that achieve the same goals with fewer resources. From using repurposed containers for composite sampling to adapting qPCR assays with locally available reagents, innovation thrives where necessity flows deepest. In one word: to better adapt to local contexts, equity shouldn't necessarily start with equal funding or resources; it should essentially start with equal imagination.

CORRESPONDANCE

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MYTH 2: "Because LMICs lack expertise, capacity building is a one-way street"

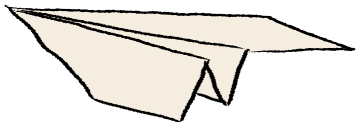
The idea that knowledge flows from high-income countries (HICs) to LMICs is outdated, and, frankly, uninspired. LMICs innovate constantly, often creating low-cost and community-driven surveillance models that match complex settings. Across Africa, Asia, and Latin America, environmental engineers, field technicians, microbiologists, and data scientists have been running (waste)water quality programs, AMR monitoring, and outbreak investigations in very challenging contexts. Just think of the complexity of surveillance in non-sewered settings, and you will get the idea. Regional labs share reagents across borders; universities train technicians on open-source pipelines;
small start-ups design affordable sensors using 3D printers. Yet, they are too often seen as "local implementers" rather than "knowledge partners". Expertise in LMICs has long existed, it just doesn't always wear a lab coat or matches fancy infrastructures. Because ultimately, surveillance systems are only as strong as the hands that sustain them, capacity building should be reimagined as capacity exchange, and equity should consider shared ownership and genuine collaboration.

MYTH 3: "Equal data generation and sharing means equity"

Sharing data is important, but only when those generating it can also interpret and use it. Otherwise, equity risks becoming data extraction. The real question is: who benefits from the data, and who has the power to act on it? Without local analytical capacity, open data can feel more like an open drain. Local supports to SARS-CoV-2 detection and real-time monitoring shouldn't have been restricted to one pathogen. LMIC contexts demand broader, more integrated systems. The same infrastructure that detects SARS-CoV-2 can also monitor enteric diseases or AMR trends, data that's actionable for everyday public health needs. In fact, supporting broader applications may be the key to sustainability: when surveillance can be tailored to reflect local priorities, it earns a real-time potential, local trust and continuity.

MYTH 4: "Technology will fix inequity"

Spoiler: it won't. Assuming that sophisticated sequencing machines, complex logistics, and high-cost consumables are the only way to make wastewater surveillance work is wrong. Sequencers, dashboards, and AI models are only as effective as the systems that sustain them.



CORRESPONDANCE

Flushing Out Myths: Rethinking Equity in Wastewater Surveillance from an LMIC Lens

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Don't get it wrong - advanced tools are extremely valuable. But technology without maintenance, continuity, local ownership, and communication is just shiny hardware sitting in labs. Sometimes, a reliable pump and a trained technician are more revolutionary than any sophisticated expensive device. True equity isn't about the next-generation sequencer or the fanciest dashboard; it's about who controls the data, who interprets it, and who decides how it's used. Long-term sustainable wastewater surveillance systems depend less on imported instruments than on local governance, consistent support, and the ability to turn data into decisions. Perhaps we should shift the question from "Who can afford the technology?" to "Who can afford to trust and use it?" Because in the end, a surveillance system that communities don't understand, or governments don't use is as good as a pipe leading nowhere.



MYTH 5: "Stakeholder engagement just requires communication"

If only it were that simple. In LMICs, engaging stakeholders, from ministries to municipalities, from utilities to communities, can feel like navigating a maze of overlapping responsibilities and shifting priorities. It's not that people don't care; it's that wastewater surveillance often falls between institutional cracks.

collecting samples in a second, data governance in another, and funding in a fourth. Add to that the constant rotation and frequent turnover of personnel, fragmented mandates, and donor-driven timelines, and you get a recipe for what I would call engagement fatigue. So yes, stakeholder engagement is difficult, but not because LMICs lack motivation or coordination skills. It's difficult because systems are often built around silos and short-term projects instead of long-term trust. Equity, in this sense, means slowing down to co-design, giving local institutions time to digest, adapt, and own the process. Real engagement isn't a meeting, it's a relationship. And like wastewater systems, relationships need maintenance to keep flowing.

Equity is the Real Early Warning Signal

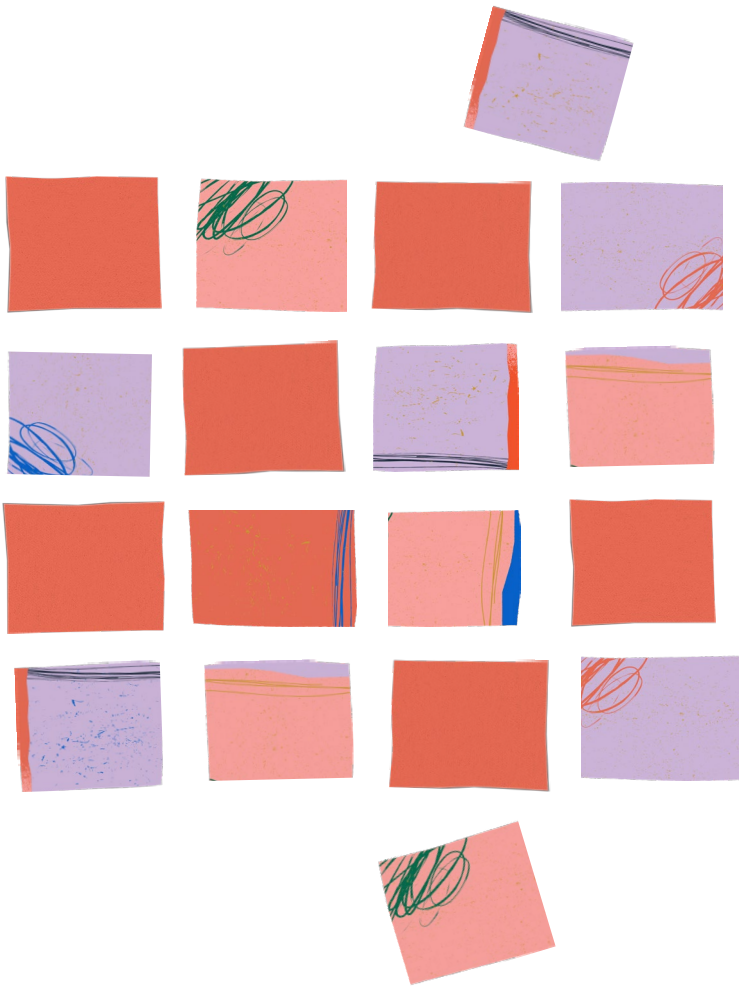
Not all myths are pure fiction. The challenge is distinguishing between what's imagined, what's imposed, and what's simply overlooked. Perhaps it's time to stop treating equity as a soft add-on and start seeing it as the hardware of sustainability. Let's build surveillance systems that don't just detect pathogens early but detect inequity even earlier. This is not only a technical mission, it's a social contract. If we truly want equitable wastewater surveillance, we need to start listening differently. Instead of prescribing models, let's ask: What's already working locally? Instead of training, perhaps it's time to co-learn.

CORRESPONDANCE

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And instead of measuring success by the number of sequencing runs, maybe we should measure it by how information reaches the communities who need it most and can be made actionable. Equity doesn't slow us down; it makes sure everyone crosses the finish line.



COMMENT

Human Habitats: Designing for a Changing Planet

Giulia Panzarella



Preface

As we rethink our relationship with nature, we must reconsider the spaces we inhabit. Human habitats, cities, towns, rural communities, are not isolated from the environment. They are embedded within ecosystems, shaped by climate, geography, and biodiversity. And in an era of accelerating change, they must evolve.
Traditional infrastructure was designed for a climate that no longer exists. Roads, buildings, drainage systems, and energy grids were built based on historical weather patterns. But today, with floods, heatwaves, and storms becoming more frequent and intense, these systems are under strain. We are no longer managing the present, we are confronting the future and **the future of human habitats lies in integration**. Urban planning must embrace nature-based solutions: green roofs, permeable surfaces, urban forests, and restored wetlands. These aren't just aesthetic choices: they regulate temperature, absorb carbon, manage water, and support biodiversity. Rural habitats, too, must be reimagined. Agricultural landscapes can become carbon sinks, and villages can serve as models of circular economies. The goal is not to retreat from modernity, but to redesign it, embedding ecological intelligence into the very fabric of our homes, neighbourhoods, and cities.
The way we shape human habitats reflects our values. Do we build for resilience, equity, and sustainability? Or do we cling to outdated models that no longer serve us?
The answer, as the physicist Giulio Boccaletti suggests, lies in conscious civic and political choices. It's not just about where we live, it's about how we choose to live.

In a world grappling with climatic, ecological, social, and economic crises, marked by deep inequalities and conflicts, how can we envision and shape a collective future? How do we design a path forward that embraces both humanity and the planet? A recently published book offers a good perspective: "The Future of Nature: Solutions for a Changing Planet" by Giulio Boccaletti, atmospheric physicist and scientific director of the Euro-Mediterranean Center on Climate Change. Known for his work on water, drought, and the natural world, Boccaletti doesn't offer technical instructions. Instead, he challenges us to shift our point of view: to see nature not as separate from us, but as deeply integrated into our modern life. Nature documentaries often depict a pristine world untouched by humans, almost alien in its purity. If an extraterrestrial observer were to base their understanding solely on these films, they might conclude that humans do not live on planet Earth.. but the reality is quite the opposite: we are everywhere, and our future is inseparable from the future of nature. Boccaletti argues that nature must be understood not just as a postcard-perfect landscape or a resource to be exploited, but as a foundational infrastructure for sustainable development. In this vision,

COMMENT

Human Habitats: Designing for a Changing Planet

Giulia Panzarella

ecosystems become tools to solve problems such as climate change, water management, disaster prevention, within a technologically advanced society. This shift is beginning to take shape in European environmental policy, where leaders are seeking a new balance between ecology and economy. During the EU autumn summit on October 23–24 in Brussels, heads of Member States debated ambitious climate goals, including the 2035 ban on internal combustion engines and the promotion of biofuels like bioGPL, renewable DME, and bioLNG. The 2040 target to reduce harmful emissions by 90% remains under scrutiny, but it reflects a broader push for systemic decarbonization and technological innovation. The urgency is undeniable: we must decarbonize our planet. Beyond political measures aimed at reducing, or ideally eliminating, human-induced CO_2 emissions to halt global warming (CO_2 traps infrared radiation, raising global temperatures - a fact derived from fundamental physics, including Planck's and Boltzmann's laws and further reaffirmed by Enrico Fermi's 1940s research on CO_2's infrared absorption and greenhouse effect), science tells us that we must also begin reabsorbing the carbon already released into the atmosphere. In this regard, the only truly scalable solution we currently have is nature itself. Marine and terrestrial ecosystems help offset emissions, but we must enhance their capacity. This means managing ecosystems for abundance and biodiversity. A new study recently published in the journal Estuarine, Coastal and Shelf Science demonstrate we are currently not moving in this direction: the warming of the sea, together with the environmental alterations caused by human activities, is causing a progressive decline of the seagrass meadows, in particular Cymodocea nodosa, in the coastal areas of the northern Adriatic. **The conservation of these habitats is considered essential for climate change mitigation** and for the protection of marine biodiversity in the Mediterranean. Science provides the tools, but it cannot act alone. Solutions must emerge from collaboration between scientists, policymakers, and citizens. It's not just about data, it's about dialogue, vision, and collective action. We need to develop integrated spatial planning strategies. Take the landslides triggered by floods across the Italian Apennines; these are not merely geological events, but symptoms of neglected forest management. Water systems present another challenge. Effective management requires more than just dams, canals, and levees. It demands an integrated approach that blends engineered solutions with natural ones, like vegetation cover, soil conservation, and watershed restoration. When ecosystems are treated as part of our vital infrastructure, they become powerful allies in building a modern, technologically advanced, and sustainable society. But this transformation doesn't happen by chance: it requires deliberate civic and political choices. In his book, Boccaletti illustrates this with compelling examples. One such example is the supply chain of the Italian Parmigiano Reggiano. dynasty. Mowed up to five times a year, these meadows yield nutrient-rich grass that enhances The cows that produce the milk for this iconic cheese feed on the cuttings from so-called "permanent meadows".

COMMENT

Human Habitats: Designing for a Changing Planet

Giulia Panzarella

These are highly biodiverse ecosystems, though not entirely natural; they were created by humans during the time of the Gonzaga milk quality, which in turn produces *Parmigiano*. What's remarkable is that these semi-natural ecosystems absorb 25% more CO_2 than conventional forage crops like alfalfa. In this case, we see a layered infrastructure: the permanent meadow, shaped by human intervention and river management, supports a forage system that feeds into a food chain, ultimately generating economic value. Decades later, researchers discovered that by strategically removing or modifying certain dams, it was possible to reconnect the river to the sea without compromising hydroelectric output. This reconfiguration allowed fish to resume their migratory patterns, revitalising the fishing industry.

In one coastal area where a dam had been removed, millions of fish returned within just two to three years. This case demonstrates a powerful lesson: had such an integrated plan been considered during the initial development phase, the communities dependent on the river could have been preserved. While not every location may offer the same flexibility, the Kootenay River story shows that ecosystem services and technological infrastructure can coexist if guided by thoughtful, forward-looking choices. **The world is full of inefficient solutions: systems and infrastructures designed without considering the full potential of nature**. **Yet, many of these can be reimagined to accommodate both the abundance of ecosystems and the demands of industrial production**.

This brings us to the present. What we are experiencing today is a shift in meteorological and climatic patterns unlike anything we've seen in the past. The statistical models that once guided infrastructure design, based on historical frequencies and intensities of rainfall, are deeper mismatch between outdated no longer sufficient. Floods, for example, are not merely the result of negligence; they reflect a systems and a rapidly changing climate. Our challenge is not just about managing the present. We must ask ourselves: what will the future look like? And how can we design infrastructure and solutions that are fit for that future? Imagining the future begins with a clear understanding of the present. The future of nature is not a distant abstraction, it is the reality to which humanity must adapt its way of life and its activities. It calls for a new mindset, one that sees nature not as a backdrop, but as a dynamic partner in shaping a resilient, equitable, and sustainable world.

COMMENT

The Risk Management Game: Building Water Reuse Governance Through Simulation and Play

Roberta Maffettone and Patricia Allgaier (JRC visiting PhD student from Oxford University)

You can't understand risk management by reading about it, you have to experience it, negotiate it, and sometimes even play it.

Europe's water future depends as much on people as on infrastructure. Achieving the objectives of the European Water Resilience Strategy to "repair the broken water cycle", promote water reuse and efficiency, and strengthen governance, requires innovative and practice-oriented approaches. Training is key to turning this vision into action. That is the goal of the European Water Academy, launched by the Joint Research Centre (JRC) to strengthen the skills, competences and partnerships needed across sectors, from agriculture to public health, to build a truly water-resilient Europe. At the JRC Water Lab, training is moving beyond traditional lectures and reports. Experiential learning formats such as labs, simulations and serious games are helping participants engage directly with real-world challenges. They promote collaboration under pressure, support collective problem solving, and help translate complex regulations into practical actions. Negotiation, cooperation and systems thinking cannot be learned from books alone; they must be practiced. Interactive learning methods such as role play, and serious games help transform abstract policies into lived experiences. Participants do not simply hear about collaboration, they experience it. They face trade-offs, share decisions and understand how different perspectives shape outcomes. This approach has proven especially valuable in the water sector, where technical, social and environmental dimensions are deeply interconnected.

By simulating real decision making, serious games help build communication, teamwork and problem-solving skills that are essential for implementing Europe's water resilience goals. Since the COVID-19 pandemic, wastewater-based surveillance has become a vital early-warning system for public health. By detecting pathogens in wastewater, authorities can track community-level trends before clinical cases emerge. Yet this field requires more than technical expertise: it depends on collaboration between laboratories, utilities, municipalities and health authorities. Hands-on exercises such as mock incident rooms where teams interpret wastewater data and coordinate responses are proving powerful tools to strengthen preparedness. They allow participants to practice rapid data interpretation, decision making and cross-agency communication in realistic scenarios. These training formats show how analytical, and governance capacities can evolve together through structured, participatory learning. Serious games are an emerging training method in water governance. They place learners inside interactive scenarios that reflect the complexity of real decisions, balancing resource management, public health and policy constraints. By taking on stakeholder roles, participants experience how different priorities, uncertainties and trade-offs interact.

An example of this approach is the **Risk Management Plan (RMP) Challenge**, a serious game developed at the JRC to simulate the process of creating a Risk Management Plan under EU

COMMENT

The Risk Management Game: Building Water Reuse Governance Through Simulation and Play

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Regulation 2020/741 on water reuse. In a two-hour facilitated session, participants take on roles such as farmer, utility operator or regulator and work together on a realistic case study. Using cards that describe hazards, exposure routes, receptors and management actions, they assess risks and decide on appropriate mitigation measures. Through discussion and negotiation, they jointly develop a draft Risk Management Plan.

The game was first played at the 4th International School on Water Reuse in Torino, where participants from seven European countries took part. The results were clear: confidence in assessing risks increased, understanding of stakeholder perspectives deepened, and overall knowledge of risk management improved significantly.

By turning regulation into experience, the RMP Challenge bridges the gap between technical knowledge and governance practice, transforming learning into collaboration.

The game embodies the spirit of the European Water Academy, which connects science, policy and operations through active learning. Alongside its technical training on monitoring, analytics and reuse operations, the Academy offers these experiential governance labs where participants can test decisions in realistic, low-risk settings.

By transforming principles into play, the Academy is helping build the shared understanding, trust and cross-sector skills needed to make the Water Resilience Strategy a reality.

So, are you ready to play with us?

Figure B| Roberta Maffettone and Patricia Allgaier presenting the serious game in Torino (on the left). Participants playing the serious game (right side). *Source: JRC*

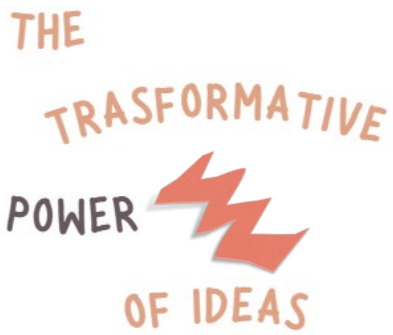




COMMENT

Destruction, Technology, Creativity

Giulia Panzarella



In October 13, 2025, the *Nobel Memorial Prize in Economic Sciences* was awarded to three distinguished researchers, i.e., **Joel Mokyr** of Northwestern University, **Philippe Aghion** affiliated with Collège de France and the London School of Economics, and **Peter Howitt** of Brown University, for their groundbreaking work explaining how innovation drives sustained economic growth. Their collective research shines a spotlight on the central role of technology and the powerful dynamic known as "creative destruction."

Creative destruction, a concept mathematically formalised by proff. Aghion and Howitt, describes the continuous process whereby new technologies and products replace outdated ones. This cycle, while disruptive to incumbents, is essential for dynamic, long-term growth. It echoes the observations of Joseph Schumpeter, who famously identified creative destruction as the "perennial gale" invigorating capitalism.

Historically, stagnation was humanity's default state: for centuries, societies largely replicated the agricultural and economic practices of their ancestors, with little transformative change. The extraordinary growth witnessed over the past 300 years, beginning with the Industrial Revolution, represents a fundamental shift. Mokyr's historical research reveals that this transformation was fueled not merely by incremental improvements, such as slightly better tools, but by the cumulative, science-based technological advancements and an openness of society to new ideas.

What makes innovation so potent, as highlighted by these laureates, is the nature of ideas themselves: they are non-rivalrous goods with the potential for infinite application. When an idea improves productivity, that innovation can diffuse widely, amplifying its impact exponentially and enabling unprecedented economic expansion.

However, innovation alone is insufficient. It requires a conducive socio-political fabric—a system of institutions, policies, and cultural openness—that nurtures and disseminates technological progress for societal benefit. The history of technology is dotted with instances where brilliant ideas failed to take root because crucial social or institutional conditions were absent. Thus, the vital question emerges: **what conditions allow ideas to flourish and meaningfully transform economies?**

Consider England in the late 18th and early 19th centuries, where its status as a maritime power and a hub of industrial activity set the stage for the first industrial revolution. Similarly, the rise of California as the epicenter of the digital revolution was no accident. The region's robust wartime industrial base, massive federal investments in military and computing technologies during and after World War II, and a vibrant cultural milieu, shaped by the social

COMMENT
Destruction, Technology, Creativity

Giulia Panzarella

upheavals of the 1960s, created fertile ground for the emergence of the semiconductor industry and the internet. **Innovation ecosystems emerge from a complex interplay of historical, political, economic, and cultural factors.** Here, strong state involvement, dynamic academic institutions, and a societal willingness to embrace new paradigms converge to spark transformative change.

Today, the question is whether we are genuinely generating new ideas or merely rediscovering existing ones. Evidence suggests that radical innovations often come not from incumbent firms entrenched in maintaining their business models, but from startups and smaller entities willing to challenge established paradigms: a phenomenon explained by Bruno Caprettini, Economist at the School of Economics and Political Science, University of St. Gallen, Switzerland. This insight has profound policy implications, highlighting the need to regulate dominant technology firms to prevent them from stifling innovation. In Italy, and globally, the challenge remains: does the environment foster such openness and creativity? Technology alone cannot resolve deep-seated social challenges. Democratic institutions must explicitly engage with innovation processes to ensure they promote equitable and sustainable development.

Unregulated technological advances risk exacerbating economic inequalities and environmental harm.

Echoing this necessity, last year's Nobel laureates, Daron Acemoglu, Simon Johnson, and James Robinso, centered their research on how political and institutional frameworks shape economic outcomes. Their work emphasises that inequality is fundamentally a political choice, influenced by policy decisions rather than technological determinism.

The 2025 Nobel laureates' findings further show that innovation's benefits tend to concentrate, often creating "winner-takes-most" markets. Without deliberate governance, technological progress may deepen disparities rather than alleviate them.

Yet, it is undeniable that over the past three centuries, human wellbeing and prosperity have improved dramatically worldwide. This progress, propelled by the relentless cycle of creative destruction, shows us the **transformative power of ideas** when paired with the right societal conditions.

As we navigate a future where artificial intelligence and other frontier technologies promise to reshape economies again, these insights offer both hope and caution. They remind us that humanity's prosperity depends not only on inventing new technologies but on cultivating the socio-political ecosystems that allow innovation to thrive inclusively and sustainably.

EU SUPER-SITES: INSIGHTS FROM THE OUTSIGHT
Madeira GIGA-Site – A European Model for Innovation in Public Health

Sara Comero, Simona Tavazzi, Natalia Głowacka, Giulia Panzarella, IT Office

The Madeira GIGA Site stands as a pioneering sentinel node within the European Union's **GLOWACON network** (Global Consortium for Wastewater and Environmental Surveillance for Public Health). It represents a powerful convergence of scientific innovation, governmental leadership, and community engagement, setting a benchmark for environmental public health action across Europe.

We extend our gratitude to the **President of the Autonomous Region of Madeira**, Miguel Albuquerque, whose presence at **Expo Osaka** for the inauguration of the "*Islands of Life*" reaffirmed Madeira's leadership as a model for environmental health action.

Designated as the **first EU SUPER-SITE** within GLOWACON, Madeira has collected the largest volume of wastewater samples in Europe, placing it at the forefront of high-resolution infectious disease monitoring. Its unique context—a vibrant mix of residents, tourists, and digital nomads on a geographically isolated island—creates both opportunities and challenges for real-time public health surveillance.

Local authorities have embraced a vision of Madeira as the "*Singapore of the EU*," actively deploying **digital health solutions, telemedicine, AI-powered surveillance, and genetic testing**. These initiatives are not theoretical; they are operationalised to serve both permanent residents and transient populations, reflecting a pragmatic, forward-looking commitment to health innovation.

As the President has noted, Madeira's transformation is rooted in **connectivity** and **technology**. Once socially and geographically isolated, the island has evolved into a **frontline innovator** aligned with EU strategic health security priorities through its integration into GLOWACON and its adoption of cutting-edge digital systems.

Madeira has also formally offered to host the **EU Water Academy**, demonstrating leadership in knowledge exchange, capacity building, and workforce development for wastewater-based epidemiology.

This commitment directly supports GLOWACON's mission of building a harmonized, interoperable, and scalable European surveillance infrastructure.

At the operational level, the **EU Joint Research Centre** coordinates standardised protocols for sampling, molecular diagnostics, metagenomics, and data integration across SUPER-SITES. Madeira's experience shows how these protocols can be effectively combined with event-based surveillance, AI analytics, and monitoring of antimicrobial resistance and environmental risks, offering a model that is both scalable and adaptable. Challenges remain: harmonising data governance, minimising technical variability, and ensuring long-term sustainability. Realising the full potential of SUPER-SITES will require sustained, multi-sectoral collaboration, bridging science, policy, and citizen engagement.

The Madeira GIGA Site exemplifies the **feasibility, impact, and strategic value** of technology-enabled environmental public health surveillance. It offers a replicable and scalable model for Europe and the world, strengthening the foundations of pandemic preparedness, health security, and resilient environmental health systems. In upcoming issues, we will give voice to Madeira's authorities and highlight how this remarkable island - an **extraordinary reservoir of biodiversity** - is also becoming a **hub of innovation bridging cultures, business, and science**, thanks to visionary leadership and a deep commitment to sustainability and collaboration.

EU SUPER-SITES: INSIGHTS FROM THE OUTSIGHT
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Figure C| From left to right: 1. Madeira beach, 2. Prof. Ricardo Jardim Gonçalves (Director of IDEAssoc), 3. Miguel Albuquerque, President of the Autonomous Region of Madeira, in the foreground, and Pedro Ramos, former Minister of Health, on the right. 4. Evening panorama.



Source: JRC

GLOWACON NEWS
Aviation Wastewater Surveillance Programme

Bernd Manfred Gawlik

The European Union has launched the GLOWACON Aviation Programme, an initiative to enhance preparedness against infectious diseases by leveraging wastewater and environmental surveillance at airports and on aircraft. Following a successful pilot exercise conducted in October 2023, which demonstrated the feasibility of coordinated cross-border wastewater monitoring for SARS-CoV-2, the fully-fledged programme now aims to scale this approach across participating international airports and airlines throughout Europe [2]. The programme focuses on sampling wastewater as close as possible to specific airplanes and flights, providing an innovative, non-intrusive method to detect viral pathogens early, including emerging variants of concern. This surveillance is designed to complement existing clinical testing by offering near real-time data to inform public health responses. By creating a network of "Super Sites" at strategic transportation hubs, GLOWACON strives to establish a standardised, pan-European monitoring system, reinforcing global efforts to detect and mitigate pandemics.

The European Commission has invited airports and related agencies to register for the programme by late October 2025, regardless of whether they participated in the pilot [3]. Importantly, the initiative seeks collaboration not only in sample collection but also in sharing data and insights.

A forthcoming comprehensive Standard Operating Procedure (SOP) will formalise guidance and best practices, evolving from the aviation sampling document originally published for SARS-CoV-2 [4].
Beyond surveillance, the programme encourages coordination.

This initiative reflects the European Union's commitment to proactive health security, harnessing technological innovation and collaborative governance to anticipate and respond swiftly to health threats in an interconnected world. As travel and mobility resume globally, GLOWACON offers a visionary blueprint for integrating environmental surveillance into comprehensive public health strategies, ensuring a safer future for all.



THE HUMAN SAMPLE
Wastewater and Environmental Surveillance under One Water–One Health Approach

Bilge Alpaslan Kocamemi, Full Professor Marmara University, Instabul

The COVID-19 pandemic began with a virus emerging at the interface between humans and wildlife and rapidly spreading across the globe through interconnected ecosystems, societies, and economies. This was a powerful reminder that the health of humans, animals, plants, and the environment deeply interconnected under the One Health approach. While the One Health concept was formally defined in 2006 by the American Veterinary Medical Association, the pandemic revealed how far the world is still far from translating this understanding into coordinated action.

What began as a health crisis quickly unfolded into social, economic, and environmental disruptions—illustrating that health challenges are rarely confined to a single sector. Responding effectively to such complex risks requires a comprehensive framework. One Health provides a powerful framework for early detection, prevention, and rapid response to zoonotic disease threats. It also addresses endemic, vector-borne, and enteric diseases, as well as non-communicable health risks, food safety, clean water, environmental pollution, and antimicrobial resistance—challenges that are intensifying under the growing pressure of climate change, one of the greatest global threats of our time.At the center of these interconnections lies water. As an essential element for all forms of life, and both a universal solvent and powerful transporter, water acts as a crucial connector within the One Health framework. Recognizing this, the One Water approach views surface water, groundwater, and wastewater as components of a single, integrated cycle. Water flows through cities, industries, and agricultural systems—used, treated, and reused, carrying nutrients, energy, contaminants, and pathogens, thereby establishing a direct linkage between ecosystems, communities, and human, animal, and environmental health.

This shared pathway illustrates a simple but profound reality: One Water and One Health are inherently connected. Strengthening water management is therefore not just a technical necessity but a strategic imperative. Taking actions aligned with the One Water–One Health approach is key to safeguarding public health, protecting ecosystems, building resilient communities, and global solidarity. key tool in operationalizing the One Water–One Health approach is Wastewater and Environmental Surveillance (WES). Emerging as a game-changing public health instrument during the COVID-19 pandemic, WES provides real-time insights into community and environmental health through the detection of viral pathogens, chemical contaminants, and other indicators present in water systems. Wastewater-based surveillance is not entirely new. It was first

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applied under the Global Polio Eradication Initiative in 1988 and later, in 2011, through the SCORE (Sewage analysis CORe group—Europe) network for monitoring illicit drug use. However, none of these earlier initiatives gained the global visibility or strategic importance achieved during the COVID-19 pandemic. Today, 72 countries are actively applying WES to support pandemic monitoring and broader public health surveillance. Building on this experience, WES now stands as a critical early warning system, enabling more timely and effective responses to potential outbreaks and strengthening the operational foundation of the One Water–One Health approach.

A major milestone in advancing this vision is the new European Union Urban Wastewater Treatment Directive (January 2025)—the first wastewater regulation to explicitly integrate the One Health approach. The directive mandates WES targeting viral and pathogenic agents, monitoring antimicrobial resistance (AMR), and identifying emerging pollutants with potential impacts on human, animal, and environmental health. Beyond disease surveillance, WES has the power to transform data into dialogue, trust, and shared action, encouraging the circular use of treated water in agriculture, strengthening food security, and building resilience against water scarcity. It also holds great promise for deployment at mass gatherings and in emergency zones affected by war, floods, or earthquakes, where rapid detection of emerging threats is crucial.

Supporting this global effort is the Global Consortium for Wastewater and Environmental Surveillance for Public Health (GLOWACON), launched in March 2024 by the European Commission Joint Research Center and EC Health Emergency Preparedness and Response Authority (HERA). GLOWACON aims to establish a global sentinel system for early detection and response to epidemic threats, connecting partners across borders and sectors through a voluntary network for real-time monitoring and data sharing. Major initiatives include the Aviation Program and the Mass Gathering Events Program, which position WES at the front line of global health security.

Türkiye has been among the early pioneers of this vision. Through the Turkish Water Institute (SUEN) under the Ministry of Agriculture and Forestry, the country has developed one of the most comprehensive national WES systems for SARS-CoV-2, now expanding its scope to include AMR, food safety, animal health, and broader One Health challenges, supported by digitalization. Türkiye has been an active member of GLOWACON, co-leading the Training and Capacity Building Working Group, and joined the first GLOWACON supersite initiative at Istanbul Airport, in close collaboration with the Ministry of Transport and Infrastructure.

THE HUMAN SAMPLE
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Since 2020, Türkiye has also played a central role in advancing this agenda globally, leading the Water Quality and Health Task Force of the World Water Council. However, despite these advances, the practical implementation of the One Water–One Health concept remains a complex and challenging task, requiring coordinated strategies, political commitment, and innovative solutions.

Despite the growing global momentum around WES and its demonstrated potential to operationalize the One Water–One Health approach, its long-term impact ultimately depends on how well systems are governed. The greatest barrier to implementation is not a lack of awareness, but a lack of connectedness. In many countries, responsibilities for water management, human health, animal health, food systems, and environmental protection remain divided across separate ministries and agencies, each operating in isolation. This fragmented structure hinders the development of coherent strategies and coordinated responses to interconnected water, health, and environmental challenges.

WES, by its very nature, sits at the intersection of multiple sectors, bridging public health, environmental management, academia, and communities. To fully realize its potential, strong cross-sectoral collaboration and integrated governance must replace fragmented approaches. Effective governance begins with building institutional capacity and fostering structured dialogue, linking ministries, scientists, multilateral organizations, and the private sector. Establishing inter-ministerial committees, joint task forces, or multi-agency platforms can create the foundation for coordinated surveillance, shared decision-making, and joint action on infectious diseases, antimicrobial resistance, chemical pollution, and contaminants of emerging concern.

There is also a need to cultivate a shared strategic vision that positions One Water–One Health at the core of resilient governance. This involves creating permanent platforms for cross-sectoral dialogue, aligning policies across sectors, and developing a roadmap for integration into national and international frameworks. By doing so, countries can move beyond fragmented initiatives toward a unified governance model that turns One Water–One Health from concept into operational reality.

In the end, the strength of this vision will depend on collective action and shared responsibility. By embracing the One Water–One Health approach and harnessing the power of WES as a unifying tool, we can shape a healthier, safer, and more resilient world for generations to come.

BRIEF COMMUNICATIONS

THE EUROPEAN WASTEWATER SURVEILLANCE DASHBOARD

The EU Wastewater Observatory for Public Health has launched a new tool: the European Wastewater Surveillance Dashboard. This innovative platform tracks the spread of infectious diseases by analysing wastewater across Europe, providing real-time insights on pathogen trends and emerging health threats to support rapid responses to disease outbreaks.



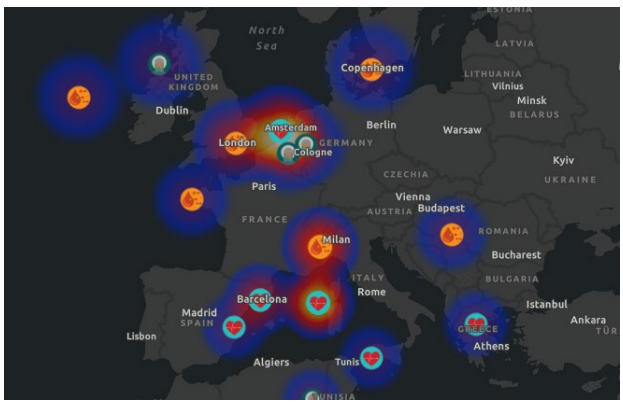
Try the tool and help us improve it - we invite you to explore the dashboard and send your feedback

ENCYCLOPAEDIA CLOACAE

Stay up-to-date with the latest information on pathogens and public health indicators in wastewater through the EU Wastewater Observatory's Encyclopaedia Cloacae.
This valuable resource supports disease detection and public health efforts across Europe.

Get involved and contribute to the growth of the Encyclopaedia by submitting publications or your organisation's logo to become an official contributor.
Our team will review and add your contributions to the platform. To contribute, please visit the "Add Your Knowledge" area.

ADVANCED HEATMAP DASHBOARD FOR WASTEWATER PATHOGEN RESEARCH MONITORING



Explore global research on pathogens in human excreta and wastewater through an interactive dashboard, featuring a dynamic heatmap of active research areas.

BRIEF COMMUNICATIONS



Are you ready to contribute to Europe's health preparedness? The European Commission is inviting expressions of interest from locations committed to using wastewater surveillance as a tool for early detection of health threats. If your site can support regular sampling—especially near transport hubs—and aligns with national or local surveillance efforts, you could become part of the European Super-Sites network. The Commission covers all sample dispatch and analysis costs and supports partnerships with key sectors.

Submit your **interest** now via the **Super-Site call** for expression of interest:
👉https://ec.europa.eu/eusurvey/runner/EU4S_SuperSites

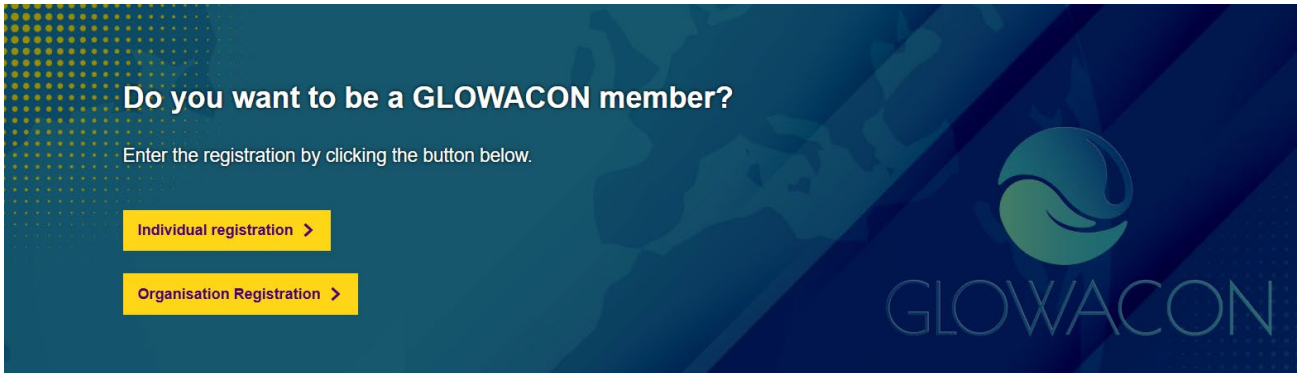
GLOWACON

GLOWACON Aviation Wastewater Surveillance Programme



With the increasing interconnectivity of global travel, the aviation sector presents a significant focus for innovative public health surveillance methods. This initiative assembles public health services, airlines, international organizations, airport authorities, academic institutions, and contracted service providers to enhance pathogen monitoring through wastewater analysis.
Through the GLOWACON Aviation Programme, we aim to address the current limitations in public health surveillance by working with the aviation sector, demonstrating the practicality and effectiveness of wastewater-based epidemiology. The insights gained will advance WES/WBE as a standard component of international public health efforts, ensuring rapid pathogen detection in a globally mobile population. Participants will not only contribute to global health safety but also benefit from cutting-edge research and data analysis opportunities, while staying in control of the derived data. Join the future of Wastewater Surveillance.

BRIEF COMMUNICATIONS

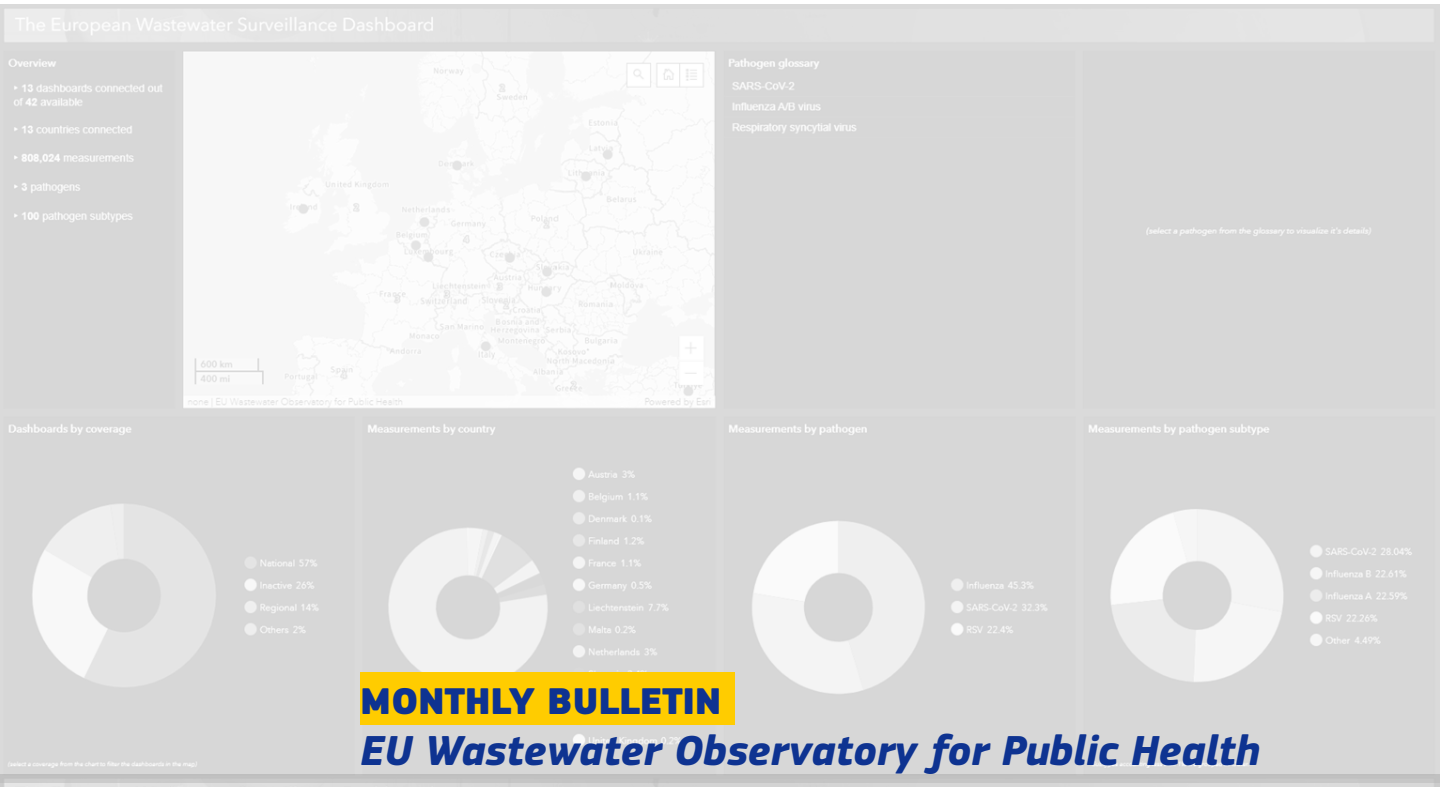


Individual registration
Organisation Registration



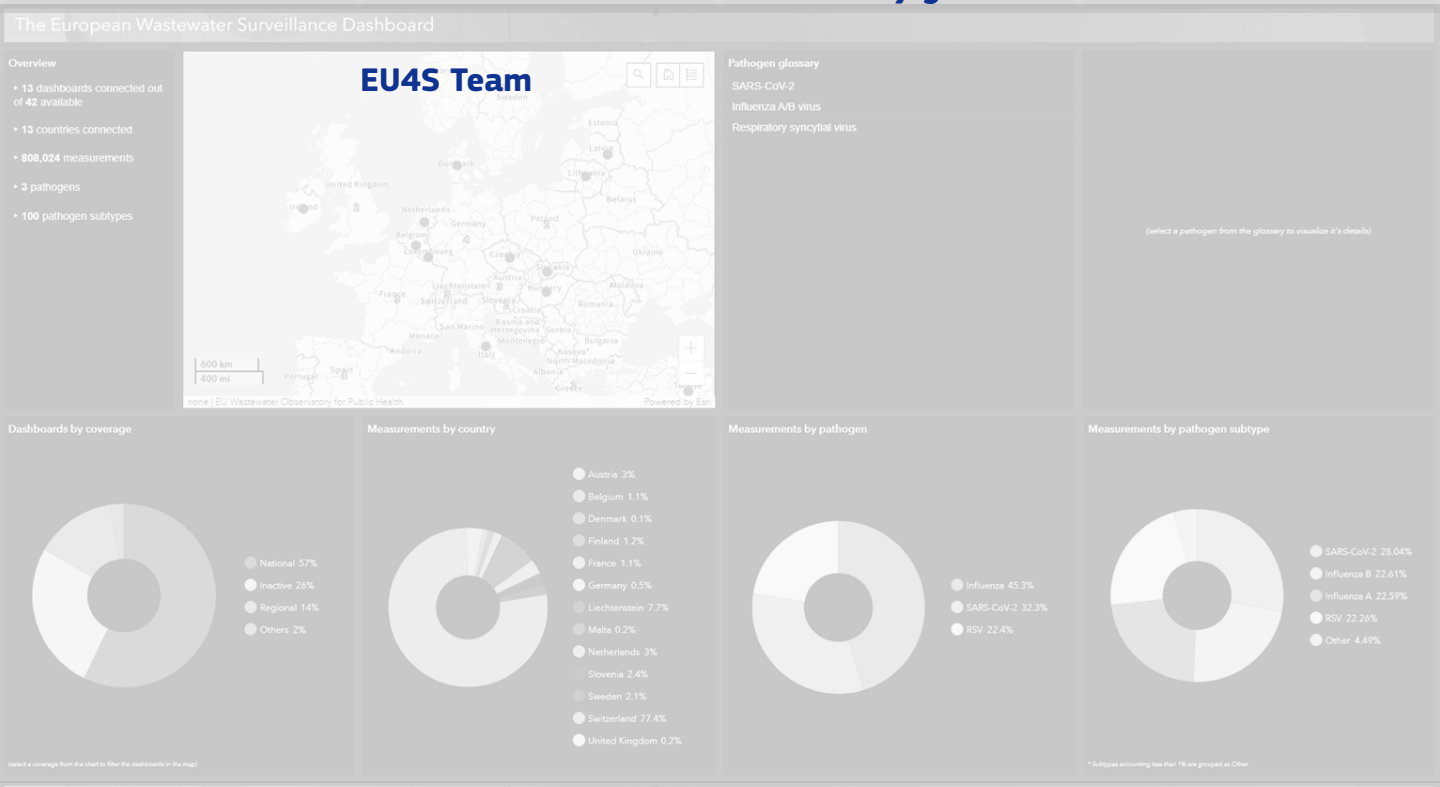
A flagship initiative to strengthen water resilience in Europe The European Union is grappling with substantial water challenges due to climate change, unsustainable practices, ageing infrastructure, and mismanagement.
In response, the European Commission has introduced the European Water Resilience Strategy, with the water efficiency first principle, featuring EUWA, the European Water Academy, as one of its flagship initiatives.
Stay tuned for further announcements.

Check out the EUWA Fact Sheet.

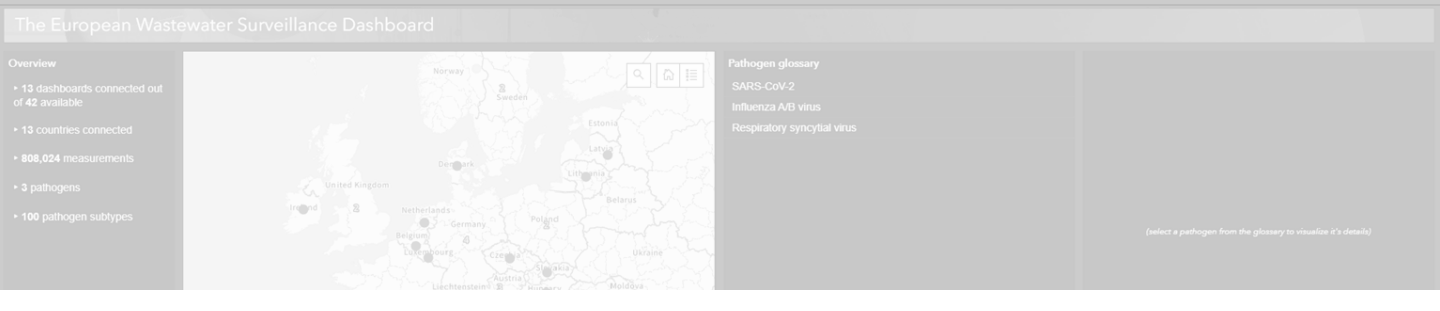


MONTHLY BULLETIN

EU Wastewater Observatory for Public Health



EU4S Team



MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

EU4S Team

EXECUTIVE SUMMARY

- **Updated SARS-CoV-2 wastewater data are available at the national level for 16 of the 27 EU countries.**
- **As of October 2025, two countries (Germany, Netherlands) have reported increasing trends in SARS-CoV-2 levels in wastewater. Finland and Hungary have reported stable levels, while Austria, Denmark and Latvia have reported decreasing levels. Viral loads are classified as moderate in Belgium, Denmark and Luxembourg.**
- **In September 2025, the XFG variant (Stratus) became the most dominant lineage of SARS-CoV-2 in Europe, quickly surpassing other Omicron subvariants. At this time, there is no evidence to suggest that XFG subvariants are more likely to cause severe illness compared to previous Omicron subvariants.**
- **Influenza and RSV continue to be monitored in Austria, Belgium, Finland, Germany, Hungary, Luxembourg and Sweden. Belgium, Finland, Ireland and Slovenia are also conducting tests for poliovirus but currently no new data have been reported. Sweden and Luxembourg conducted surveillance for enteric viruses until mid-2024.**
- **Dengue, chikungunya, Zika, and yellow fever are emerging as significant arboviral diseases of growing concern. However, at present, only sporadic wastewater surveillance data are available.**
- **The EU Wastewater Observatory for Public Health is keenly following these developments and will integrate new insights in forthcoming bulletins, with input from the EU Super-Sites.**

This bulletin offers insights derived from the latest wastewater surveillance data primarily obtained from EU national dashboards and websites owned and operated by National Health Authorities. Additionally, data from existing regional/local, and non-EU sources are included. The most recent retrieval was performed on *31st October 2025*.

Wastewater surveillance analyses samples to reveal community health and environmental trends, enhancing traditional public health monitoring; the EU Wastewater Observatory collects and shares this data to support decision-making and pandemic preparedness, tracking viruses like SARS-CoV-2, avian flu, mpox, RSV, and influenza. In June 2025, the European Commission's Health Emergency Preparedness and Response Authority (HERA), in partnership with the Joint Research Centre (JRC), initiated the sample collection phase of the European Super-Sites initiative, implemented through a HaDEA (European Health and Digital Executive Agency) framework contract. This network of strategically located sites monitors

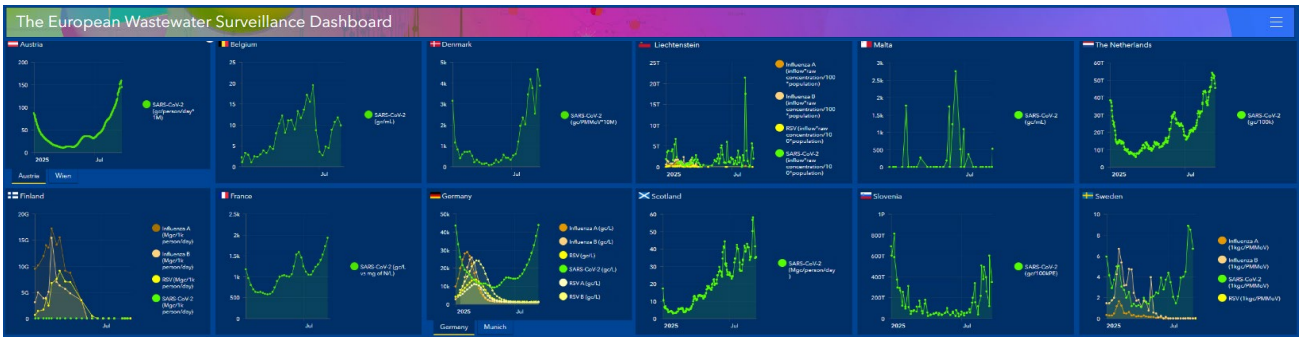
MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

wastewater near transport hubs to detect emerging health threats. As part of the Global Wastewater Sentinel System, the initiative enhances early warning capabilities and supports public health decision-making across the EU and beyond.

SARS-CoV-2 surveillance

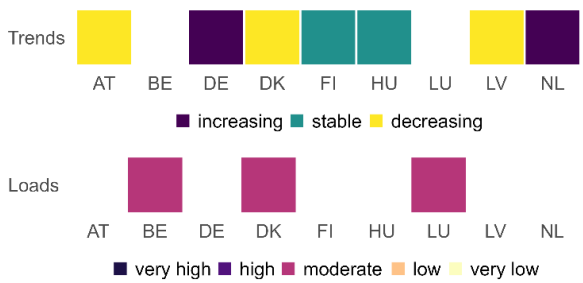
Figure 1. Overview of the European Wastewater Surveillance Dashboard: near real-time wastewater data on measured pathogens for connected countries, providing early warnings and rapid responses to protect public health across Europe.



Source: The European Wastewater Surveillance Dashboard.

A summary of SARS-CoV-2 trends and viral loads is provided (Figure 2) for various EU countries. Figure 2 includes only those countries for which data on viral loads and trends is available. If information is missing, it indicates that either the national trend or viral load has not been specified, or the data has not been updated to reflect the current month.

Figure 2. Viral loads and trends of SARS-CoV-2 across EU countries.



Source: *EC-JRC*

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

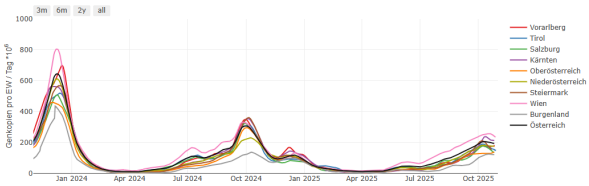
AUSTRIA

Last data update: 29 October 2025

Austria's monitoring of SARS-CoV-2 involves 48 strategically chosen sewage treatment plants, encompassing over 58% of the country's population.

SARS-CoV-2: In week 44, a slight decrease in viral loads has been observed, keeping the trend indicator below 1. In week 42, the variant XFG was the most prevalent, including its sub-lineages, which accounted for approximately 80% of reported cases.

Figure 3. Person-weighted trends in the federal states (and Austria as a whole).



Source*: Abwassermonitoring*.

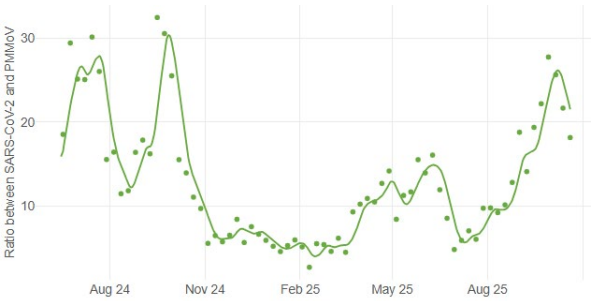
BELGIUM

Last data update: 20 October 2025

The surveillance program encompasses 38% of the Belgian population, involving the collection of wastewater samples once a week from influents of 30 wastewater treatment plants distributed across Belgium, mainly in areas of high population density.

SARS-CoV-2: In week 43, the national level of SARS-CoV-2 was classified as moderate.

concentration at a national level.



Source: *Wastewater surveillance Respiratory viruses Sciensano Belgium*.

CZECH REPUBLIC

Last data update: 29 September 2025

Systematic monitoring of wastewater for the presence of SARS-CoV-2 has been conducted since February 2023 at five wastewater treatment plants connected to the largest cities in the Czech Republic and Václav Havel Airport in Ruzyna. The monitored collection points cover approximately 21% of the Czech Republic's population.

SARS-CoV-2: The dashboard, updated to week 40, shows SARS-CoV-2 loads in each treatment plant separately, while at the national level, the presence of the viral load is given as a percentage of positive detections. Currently, this information is only available in Czech.

MONTHLY BULLETIN

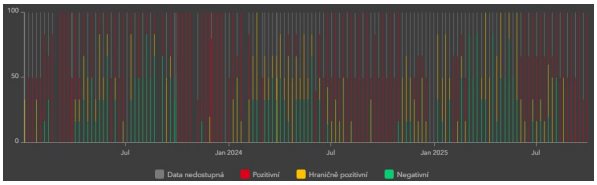
EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

CZECH REPUBLIC

Last data update: 19 August 2025

Figure 5. Presence of SARS-CoV-2 viral load in monitored wastewater.



Source: *Monitoring SARS-CoV-2 concentrations in wastewater*

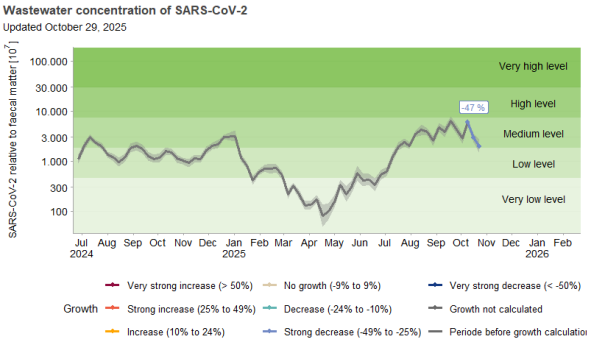
DENMARK

Last data update: 26 October 2025

Denmark's national surveillance of SARS-CoV-2 in wastewater covers 49% of the population. Twice a week, samples are collected from 29 wastewater treatment plants. The results from the wastewater are compiled for the entire country, as well as for the five regions and each specific sampling location.

SARS-CoV-2: In week 43, the concentration of SARS-CoV-2 in wastewater across the country was at a medium level. The national growth rate, reflecting the average weekly growth over the past three weeks, indicates a strong decrease in the SARS-CoV-2 wastewater concentration (-47%).

Figure 6. Concentration of SARS-CoV-2 in the wastewater, aggregate for all the sampling sites in Denmark.



Source: *National wastewater surveillance of SARS-CoV-2*.

FINLAND

Last data update: 27 October 2025

Various wastewater treatment plants across different regions of Finland actively contribute to the monitoring efforts. Presently, about 44% of the wastewater generated by the Finnish population is included in the wastewater monitoring initiative.

SARS-CoV-2: The results of monitoring coronavirus in wastewater are available at both local and national levels, updated until week 44. The method of determination changed at the end of 2024.

A statistical model estimates the trends, which reveals a stable trend in recent weeks.

MONTHLY BULLETIN

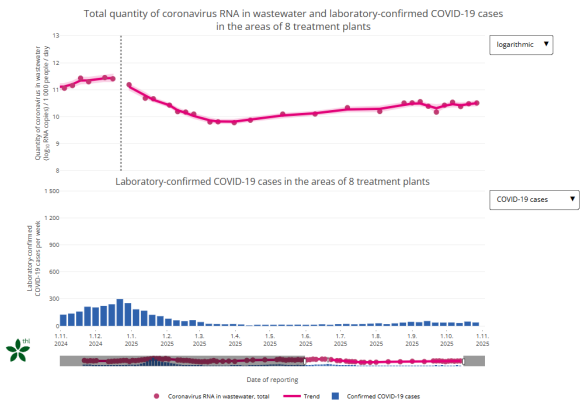
EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

FINLAND

Last data update: 27 October 2025

Figure 7. Total quantity of coronavirus in wastewater and laboratory-confirmed COVID-19 cases, in the areas of 8 treatment plants.



Source: *Coronavirus wastewater monitoring weekly report*.

FRANCE

Last data update: 19 October 2025

The national wastewater monitoring system, SUM'Eau, was launched by the General Directorate of Health and the Directorate for Water and Biodiversity, with expert support from health agencies. The data allows for estimating the SARS-CoV-2 circulation trend in 54 treatment plants across 12 administrative regions.

SARS-CoV-2: The data has been updated to week 37 and is accessible in the form of CSV files. The SARS-CoV-2 monitoring indicators in wastewater are computed as the ratio between the viral concentration of SARS-CoV-2 and the concentration of ammoniacal nitrogen. The overall national indicator is also available.

Figure 8. Structure of the CSV file available containing Sars-Cov-2 wastewater surveillance of data.



semaine	AMIENS	ANGERS LA BAUMETTE	ANGLET (PONT DE L'AVEUGLE No2)
2025-S42	8729.83097429892	3122.49999745027	1015.37851071159

Source: *Data Explorer data.gouv.fr*.

GERMANY

Last data update: 08 October 2025

The viral load of SARS-CoV-2 in wastewater is currently assessed at 169 wastewater treatment plants, providing a coverage for approximately 32% of the population. The number of reporting locations fluctuates over time.

SARS-CoV-2: During the past 4 weeks (reporting week 41), average SARS-CoV-2 viral load measured at wastewater treatment plants has increased.

Figure 9. Mean viral load in wastewater.



Source: *Infection Radar*.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

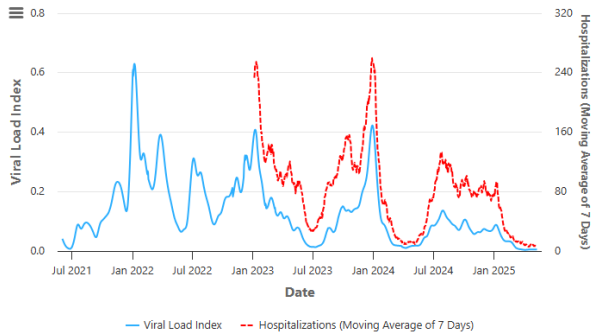
GREECE

Last data update: 21 September 2025

The dashboard presents data from the National Wastewater Epidemiology Network (NWEN), which monitors SARS-CoV-2 and other respiratory viruses in urban wastewater across 10 regions in Greece, covering 70% of the population. It visualises weekly viral load trends at both city and national levels using maps and charts, based on frequent sampling and analysis by coordinated laboratories. The dashboard also includes data on pharmaceuticals and chemicals found in wastewater, offering a comprehensive tool for public health surveillance and early warning.

SARS-CoV-2: During week 38, the national viral load was very low, showing a -0.8% change compared to the previous week, which indicates a stable trend. Among the 8 monitored Wastewater Treatment Plants, 2 displayed an increasing trend, 5 showed a decreasing trend, and 1 remained stable.

Figure 10. National index variation in viral load levels.



Source: *Wastewater surveillance dashboard*.

HUNGARY

Last data update: 26 October 2025

The data for wastewater monitoring in Hungary are obtained from three wastewater plants serving the entire area of Budapest and a portion of the agglomeration. Additionally, 19 wastewaters from five other agglomeration settlements are examined as a combined sample. Collectively, these analyses encompass over 40% of the domestic population.

The weekly report is currently only available in Hungarian.

SARS-CoV-2: During week 43, the national average concentration of SARS-CoV-2 in wastewater was stable. Specifically, 2 areas experienced an upward trend, 19 areas displayed stagnation, and a decrease was noted in one region. From week 37, wastewater analysis detected the XFG variant as predominant, alongside occasional findings of Nimbus, LP.8.1, JN.1, and other X-recombinants.

Figure 11. National, population-weighted average of SARS-CoV-2 copy number measured in wastewater. The red dots show the raw data, the grey band indicates the 95% confidence interval.



Source: *National Center for Public Health and Pharmacy*

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

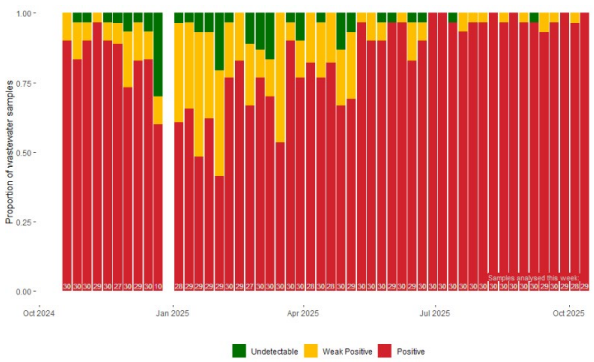
SARS-CoV-2 surveillance

IRELAND

Last data update: 18 October 2025

Since January 2023, Ireland's National Wastewater Surveillance Programme has been conducting weekly sampling in 30 wastewater catchment areas. The 30 catchment areas, covering 70% of the population connected to public wastewater treatment facilities, are distributed across all regions of Ireland.
SARS-CoV-2: In week 42, samples were collected and analysed from 28 out of 30 catchment areas. SARS-CoV-2 RNA was detected in 100% of these samples. Among the 28 catchment areas where comparisons with the previous week were possible, the viral load remained stable (within +/- 10%) in 26 areas, while two areas experienced a decrease of 10% or more.

Figure 12. Wastewater SARS-CoV-2 detections in Ireland. Week 26 2025 (22/06/2025 to 28/06/2025).



Source: *Report week 35.*

ITALY

Last data update: 04 April 2025

Wastewater monitoring in Italy is conducted through Flash Surveys, comprising monthly sampling campaigns at various locations across the country over a short duration.
SARS-CoV-2: The monitoring results are usually reported every two months, with an average delay of a few months from sample collection to reporting. The latest update is from April 2025, where a total of 107 wastewater samples were collected from 16 regions and 2 autonomous provinces (A.P.). Mutations typical of the Omicron JN.1 lineage and its sublineages were identified through Sanger sequencing in three regions, while sequencing data were unavailable for the others.

LATVIA

Last data update: 30 October 2025

The national monitoring program in Latvia focuses on identifying the presence of SARS-CoV-2 in wastewater across municipalities. Twice a week, untreated wastewater samples are gathered from various facilities within municipal centralised wastewater collection systems. The weekly is currently only available in Latvian.
SARS-CoV-2: In week 44 the overall trend of RNA concentration in Latvia was moderately decreasing. Detailed reports on the prevalent variants are available for each wastewater facility, revealing that Omicron KP.3 sub-lineage was the most common variant in all locations during week 44.

MONTHLY BULLETIN

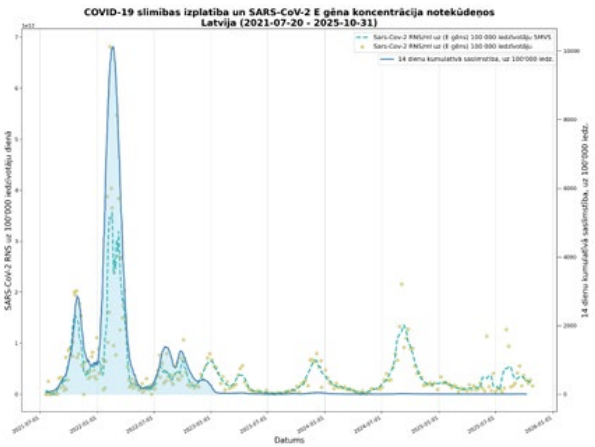
EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

LATVIA

Last data update: : 30 October 2025

Figure 13. Spread of COVID-19 and detection of SARS-CoV-2 in wastewater.



Source: Wastewater monitoring – Latvia.

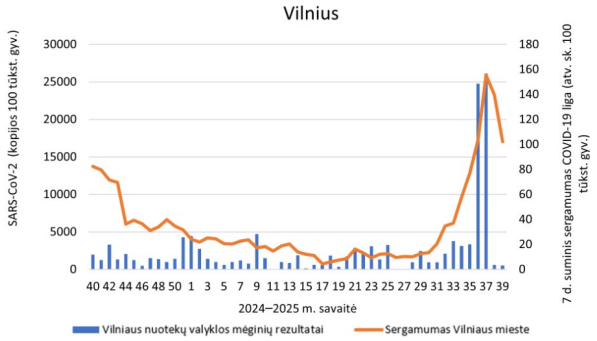
LITHUANIA

Last data update: 28 September 2025

Lithuania has implemented a wastewater surveillance system to detect the SARS-CoV-2 virus and its variants since November 2022. Wastewater samples are collected weekly from three treatment plants.

SARS-CoV-2: Three wastewater treatment facilities are currently under surveillance. The website offers data for each municipality individually, and the latest updated data is from week 39.

Figure 14. Vilnius City Wastewater Treatment Plant (copies/ 100 000 pop.)



Source: Monitoring results 2024-2025 - National public health centre under the Ministry of Health.

LUXEMBOURG

Last data update: 20 October 2025

The program for monitoring the evolution of COVID-19 in wastewater in Luxembourg involves the collection of samples from eight strategically located wastewater treatment plants, covering five cantons and approximately 67% of the country's population. Results are published in fortnightly reports.

SARS-CoV-2: SARS-CoV-2 concentrations are reported at both the national and individual plant levels. In week 43, the national levels of SARS-CoV-2 are moderate

MONTHLY BULLETIN

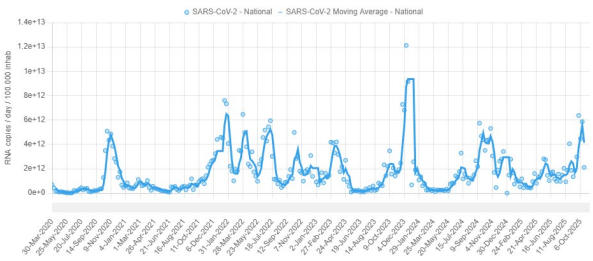
EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

LUXEMBOURG

Last data update: 20 October 2025

Figure 15. SARS-CoV-2 national levels (RNA copies/day/100.000 inhab).



Source: SARS-CoV-2 – MicrObs.

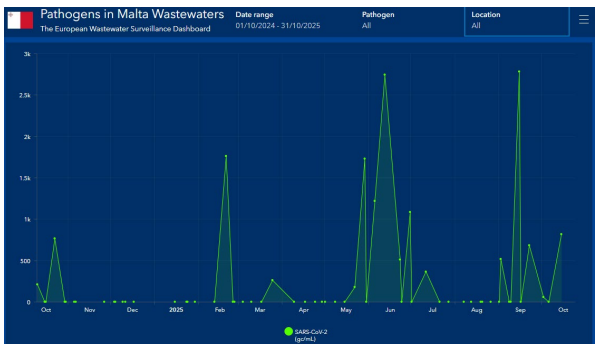
MALTA

Last data update: 24 October 2025

Malta does not currently have a dedicated national dashboard for monitoring SARS-CoV-2 levels in wastewater. However, the Ministry for Health and Active Ageing in Malta regularly submits updates on these measurements to the EU Wastewater Observatory for Public Health. Samples are collected from three wastewater treatment plants.

SARS-CoV-2: SARS-CoV-2 concentrations are reported at both national and individual plant levels. The latest results are updated to week 43.

Figure 16. Malta Dashboard.



Source: Pathogens in Malta wastewaters.

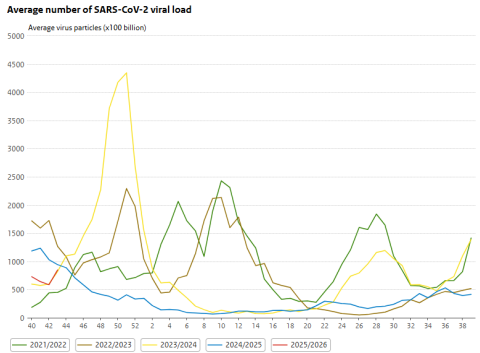
NETHERLANDS

Last data update: 26 October 2025

Wastewater samples are gathered at wastewater treatment plants throughout the Netherlands, ensuring comprehensive coverage.

SARS-CoV-2: During week 43, the national average levels of SARS-CoV-2 in wastewater increased (+45%) compared to the previous week. Multiple sub-variants of JN.1 are currently being detected in wastewater, with subvariant XFG being the most common.

Figure 17. Netherlands. Average virus load (x100 billion) over time per 100,000 inhabitants.



Source: Weekly coronavirus SARS-CoV-2 figures | RIVM.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

POLAND

Last data update: 29 January 2025

The dashboard is supported by a network of voluntary laboratories initiated by the National Institute of Public Health – National Institute of Hygiene (NIZP PZH - PIB). This network includes Provincial Sanitary and Epidemiological Stations, as well as local water supply and sewage companies. The SARS-CoV-2 wastewater data are normalised using PMMoV and reported for four cities, encompassing a total of 15 WWTPs.

SARS-CoV-2: Aggregated city data and specific site data are reported in plots. SARS-CoV-2 mutations detected in wastewater samples are reported for each city. The latest update is from the end of January 2025, and no further updates have been provided since the last monthly bulletins.

SLOVAKIA

Last data update: 16 March 2025

Wastewater surveillance for the presence of SARS-CoV-2 is conducted on a weekly basis in each region of Slovakia, with over 55 wastewater treatment plants being part of the project. Throughout the year, the monitoring intensity is adjusted based on the prevailing epidemiological situation.

SARS-CoV-2: The latest update is from March 2025, and no further updates have been provided since the last monthly bulletin.

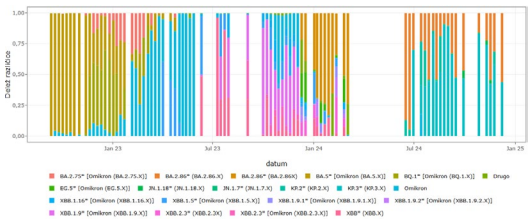
SLOVENIA

Last data update: 25 August 2025

From autumn 2022 and throughout 2024, Slovenia has been consistently monitoring SARS-CoV-2 in wastewater, using samples from 16 waste treatment plants across the nation. This monitoring network reaches about 34% of the Slovenian population and involves weekly sampling.

SARS-CoV-2: The dashboard displays SARS-CoV-2 wastewater monitoring data, including the percentage of circulating virus variants, for each individual wastewater treatment plant. The latest update is from August 2025, and no further updates have been provided since the last monthly bulletin. Results can be accessed via the Slovenian open data portal.

Figure 18. Mutation rates of the SARS-CoV-2 genome.



Source: C19-water.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

SPAIN

Last data update: 22 June 2024

Wastewater surveillance in Spain involves the examination of the presence and evolution of the genetic material of SARS-CoV-2 RNA and the predominant variants in all treatment plants, totaling 54 facilities distributed throughout the country.

SARS-CoV-2: The summary report and dashboard have not been updated since week 25 of 2024.

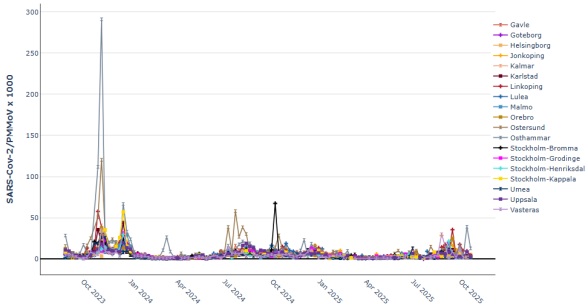
SWEDEN

Last data update: 20 October 2025

The dashboard assesses SARS-CoV-2 levels in wastewater across various Swedish cities. For weekly SARS-CoV-2 monitoring, Sweden's public health agency collaborates with SEEC-SLU (the Swedish University of Agricultural Sciences), providing data covering 43% of the Swedish population with the broadest geographic reach across the country.

SARS-CoV-2: The levels of SARS-CoV-2 in wastewater are visualised using interactive plots. The SEEC-SLU project provides weekly data updated to week 43, which can also be downloaded as a dataset.

Figure 19. PMMoV normalised levels of SARS-CoV-2 virus in multiple cities across Sweden.



Source: SARS-CoV-2 quantification | The Swedish Pathogens Portal.

REGIONAL EU DATA

The EU member states listed below furnish local or regional data regarding the surveillance of SARS-CoV-2 in wastewater for specific areas within their respective countries. The accompanying list includes accessible databases, along with their latest updates.

Austria

Vienna

Weekly wastewater analysis from the Vienna sewage treatment plant is part of the 'CSI Wastewater' project. The graphic shows the weekly occurrence of SARS-CoV-2. Data is updated to week 43, 2025.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

REGIONAL EU DATA

Germany

Berlin

The dashboard presents wastewater monitoring data from 4 treatment plants in Berlin. It shows SARS-CoV-2 data results in gene copies per litre of wastewater sampled from each plant. The analysis also includes
circulating virus variants in Berlin, presented as a percentage of the total load. In week 43, the trends indicate a decrease in virus levels in three plants and an increase in one plant.

Bavaria

The dashboard shows wastewater monitoring data from the German state of Bavaria. It includes the SARS-CoV-2 trend, viral loads and variants in wastewater from thirty municipalities within Bavaria. In week 44, measurements were conducted at 21 locations and the virus activity for SARS-CoV-2 is high.

Poland

The website provides surveillance data on the SARS-CoV-2 virus detected in sewage within the Warsaw agglomeration. The findings are presented through graphs that illustrate the normalised number of viral copies at four specific locations within the sewage network. The data is updated weekly, with results reflecting data up to week 43. An increase in virus levels was recorded at all sampling points since August A particular increase in virus content was observed in mid-October at one location, marking the year's highest level, followed by a sharp decline at the end of the month.

Spain

Madrid

The website presents visualisations of SARS-CoV-2 data in wastewater for the Community of Madrid. A map displays the progression of SARS-CoV-2 levels in wastewater across various municipalities and districts, but it is not updated monthly; the last update was in April 2025. Reports are available in Spanish.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

EU NEIGHBOURING COUNTRIES

This section provides wastewater monitoring details from neighbouring countries of the EU participating as observers in the EU National Contact Points (NCP) Mechanism. Links to national or local dashboards/websites and information about the latest updates are included.

Türkiye

Last data update: 13 July 2025

This website provides information on wastewater monitoring data in Türkiye, explicitly reporting SARS-CoV-2 viral load results. PDF reports are published for various regions (23 pilot cities), while Istanbul has its dedicated report. Wastewater samples are taken weekly from the city of Istanbul and bi-weekly from the other 23 cities, providing a coverage for approximately 41% of the population.

SARS-CoV-2: The latest update is from July 2025, and no further updates have been provided since the last monthly bulletin.

Figure 20. SARS-CoV-2 viral load distribution map, Istanbul city.



Source: COVID-19 Türkiye.

OTHERS

This section contains details on wastewater monitoring from non-EU countries. It includes links to national or local dashboards/websites and the latest updates, organised by continent.

EUROPE

Scotland

Last data update: 22 October 2025

The program has been operating in Scotland since June 2020. Each week, Scottish Water collects approximately 150-200 sewage samples from 115 wastewater treatment facilities across the country. These samples are then sent to NHS Lothian for testing (carried out by the Scottish Environmental Protection Agency until end July 2024).

SARS-CoV-2: Data is available at the national level, as well as for NHS Health Boards and local authority areas. The dashboard is updated to week 43.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

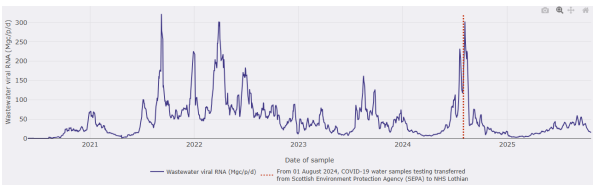
OTHERS

EUROPE

Scotland

Last data update: 22 October 2025

Figure 21. Seven-day average national trend in COVID-19 wastewater.



Source: COVID-19 & Respiratory Surveillance.

Switzerland

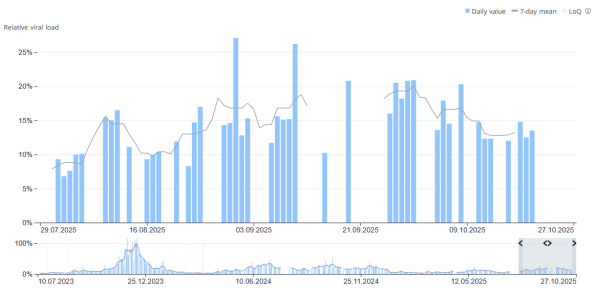
Last data update: 20 October 2025

The information published in the dashboard is based on data from Switzerland's national wastewater monitoring program, which has been tracking the viral load of SARS-CoV-2 since January 2022. As of January 2025, wastewater monitoring has been streamlined to include only 10 WWTPs in Switzerland, covering almost a quarter of the Swiss resident population, and one WWTP in the Principality of Liechtenstein. Additionally, variant monitoring is carried out at six WWTPs from December 2024.

SARS-CoV-2: Each region reports both the relative and absolute viral loads, as well as trends and variants.

During week 43, it was observed that viral loads remained stable in all 11 regions. The most frequently identified SARS-CoV-2 variant (latest update on 04.10.2025) was XFG in all the regions.

Figure 22. Relative viral load of SARS-CoV-2 at a selected location (Lugano).



Source: IDD - Infectious Diseases Dashboard.

AFRICA

South Africa

Last data update: 25 October 2025

The National Institute for Communicable Diseases (NICD) Centre for Vaccines and Immunology provides updated information on SARS-CoV-2 levels in wastewater across 14 districts in South Africa.

SARS-CoV-2: The dashboard presents SARS-CoV-2 levels both nationally and by district. Data is updated up to week 42. As of the beginning of September 2025, the most prevalent sub variant was XFG.X at 89%.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

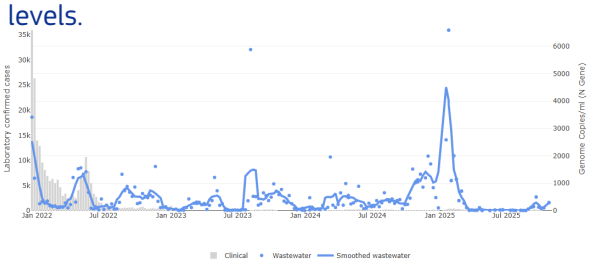
OTHERS

AFRICA

South Africa

Last data update: 25 October 2025

Figure 23. National SARS-CoV-2 wastewater levels.



Source: https://wastewater.nicd.ac.za/.

ASIA

India

Last data update: 19 October 2025

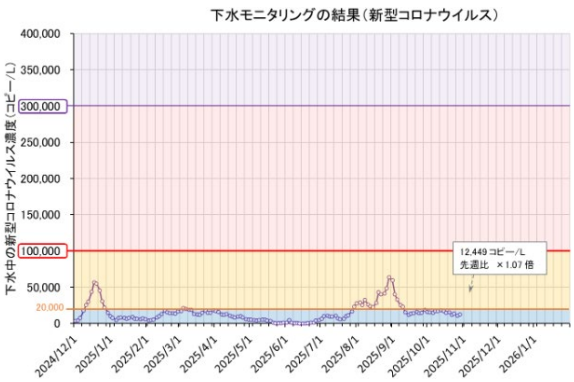
The dashboard displays the concentration of SARS-CoV-2 RNA copies per litre of wastewater in five major cities of Gujarat, India. Samples are collected from 40 wastewater treatment plants. The Lineage Prevalence section is also available. SARS-CoV-2: The latest update on SARS-CoV-2 concentration in wastewater is from October 2025 (week 42).However, information on variants has not been updated since July 2024.

Japan

Last data update: 30 October 2025

The information presented on this website reveals the outcomes of regular assessments of COVID concentration in the wastewater of Komatsu City.
SARS-CoV-2: During week 44, SARS-CoV-2 levels in wastewater were low, showing a stable trend (1.07 times) compared to the previous week

Figure 24. Virus concentration in Komatsu City expressed in copies/liter.



Source: Sewage Monitoring/Komatsu City Website.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

OTHERS

OCEANIA

Australia

Western Australia

Last data update: 24 October 2025

The dashboard presents an analysis of SARS-CoV-2 levels in the Perth Metropolitan area of Western Australia, starting from July 2022. As of August 2024, wastewater samples are collected weekly from three metropolitan wastewater treatment plants that serve approximately 79% of the city's population.

The concentrations of the virus genome in wastewater are shown as a 14-day moving average and have been updated to week 38. Omicron sub-lineage NB.1.8.X predominated across all three metropolitan wastewater sites (47.2%), followed by BA.3.2.X (20%).

New South Wales Australia

Last data update: 25 October 2025

The New South Wales Sewage Surveillance Program collects samples from 4 sites in Greater Sydney and Newcastle every week to monitor COVID-19 concentrations. The results are published in weekly reports. During week 43, gene concentrations per 1,000 people have been low in all the catchment areas.

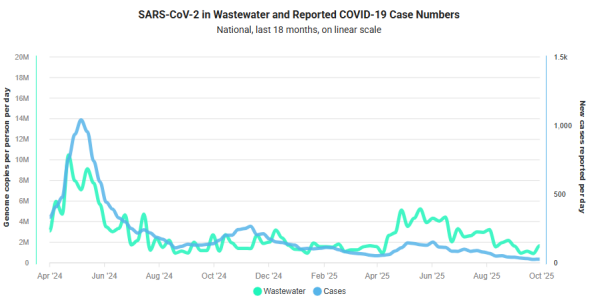
New Zealand

Last data update: 26 October 2025

The dashboard was launched in 2020 to help the public monitor potential COVID-19 risks in their local areas using easy-to-understand data visualisations. Starting in August 2024, data are updated every two weeks, and information on variants is updated at least once a month. Sites and frequency may change over time. Beginning in July 2024, variant analysis is conducted across all wastewater sites.

SARS-CoV-2: SARS-CoV-2 concentrations in wastewater are being monitored at both national and regional levels. In week 43, 10 sites covering 38% of the population tested positive for SARS-CoV-2. In week 42, the dominant strain were the NB.1.8.1 (22.2%), MC.10.2.1 (21.8) and XFG (21.7).. Trends are visually represented on a map.

Figure 25. Average SARS-CoV-2 genome copies detected per person per day in wastewater for New Zealand, along with the reported case numbers.



Source: *ESR Wastewater Surveillance*.

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EU Wastewater Observatory for Public Health

SARS-CoV-2 surveillance

OTHERS

NORTH AMERICA

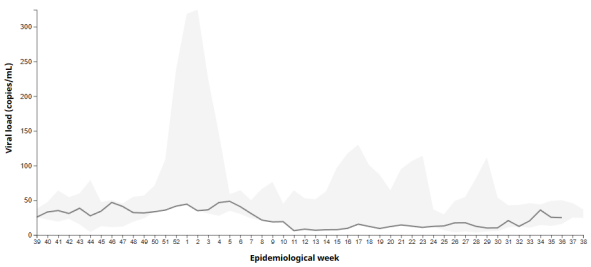
Canada

Last data update: 19 October 2025

The dashboard provides information on the levels of COVID-19 and its variants detected in the wastewater from various communities across Canada. Currently, 44 sites submit sewage samples for processing, representing 31.2% of the population. Additionally, the dashboard includes links to 23 Canadian wastewater COVID-19 surveillance dashboards and websites, which are managed by provincial, territorial, and academic partners across the country.

SARS-CoV-2: In week 42, the COVID-19 activity level in Canada was classified as moderate. Out of the 44 tested sites, 11 showed a high activity level, 14 showed a moderate activity level, 9 showed a low activity level, 6 had no detection and 3 were not available. At the end of September 2025 the dominant variant identified was XFG (73.5%).

Figure 26. Trend graph of 7-day rolling average of COVID-19 viral load in wastewater.



Source: Wastewater monitoring dashboard – Canada.ca

United States

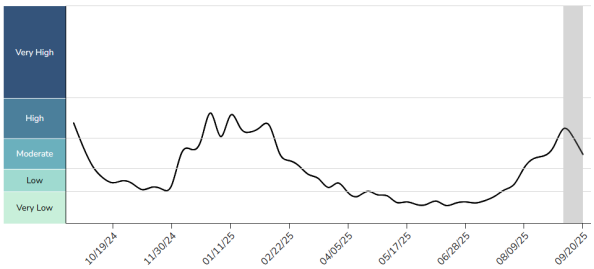
Last data update: 20 September 2025

The National Wastewater Surveillance System (NWSS) collects and analyses wastewater samples on a weekly basis. This monitoring is conducted in all 50 states, 7 territories, and some tribal communities. Over the past two months, 1,341 sampling sites have reported data to the NWSS, representing approximately 44% of the U.S. population.

SARS-CoV-2: During week 38, the national viral activity level for COVID-19 in wastewater was moderate.

The predominant variant during this week was XFG, which accounted for 83% of the total abundance.

Figure 27. COVID-19 Wastewater Viral Activity Level Over 1 Year Time.



Source: COVID-19 Wastewater Data – National Trends | NWSS | CDC.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

Mpox surveillance

This section provides a comprehensive overview of recent mpox outbreaks with a focus on wastewater surveillance and related news from around the world.

On September 5, 2025, the World Health Organization (WHO) officially declared the end of the global Mpox emergency, following sustained reductions in case numbers and mortality rates, particularly in the Democratic Republic of Congo, Burundi, Sierra Leone, and Uganda. Despite this milestone, Mpox remains a significant public health concern, especially in Africa, which continues to account for over 90% of reported cases. According to the WHO's latest situation report (dated September 19), a total of 3,780 confirmed cases and 15 deaths were recorded globally in August. Both Clade I and Clade IIb strains remain in circulation.

Mpox transmission in Europe remains low but persistent. Most cases continue to involve Clade IIb. In recent months, several EU/EEA countries—including Germany, France, Belgium, Sweden, and Ireland—have reported isolated Clade I cases, all linked to travel from endemic regions in Africa. The UK confirmed 16 Clade I cases and over 580 Clade IIb cases since 2023, with no evidence of sustained community transmission of Clade I. Berlin experienced a rise in Clade IIb cases earlier in the year, prompting renewed public health outreach. Overall, although Europe has seen some cases, the situation remains under control due to ongoing surveillance and preventive measures.

Mpox wastewater surveillance

This section provides information on mpox wastewater surveillance derived from the latest wastewater surveillance data primarily obtained from EU national dashboards and websites owned and operated by National Health Authorities.

EU Countries

Spain

Within the framework of the Spanish Wastewater Surveillance Programme, in December 2024 and January 2025, 10 different sampling points were assessed and 100% of them tested negative for mpox by RT-qPCR assays. The team is still working on returning to take weekly samples on 66 sampling points, due to some funding matters, and assess again routine SARS-CoV-2 samples, adding mpox, enterovirus and poliovirus to the monitored agents of interest list.

Non-EU Countries

Canada

The wastewater dashboard summarises the consistent detection of mpox virus signals in wastewater samples from selected cities across Canada. Signal interpretation is based on data from only four weeks of testing. Recent data from week 41 indicate no detection, or detection for just one week in the past four weeks, across 5 cities. In contrast, there was consistent detection, defined as detection of mpox for more than 2 weeks in the past 4 weeks, in one city.

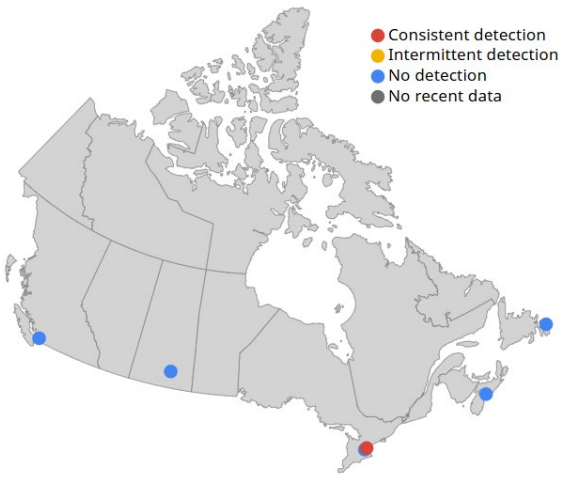
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EU Wastewater Observatory for Public Health

Mpox surveillance

Canada

Figure 28. Recent mpox detection in wastewater in cities across Canada.

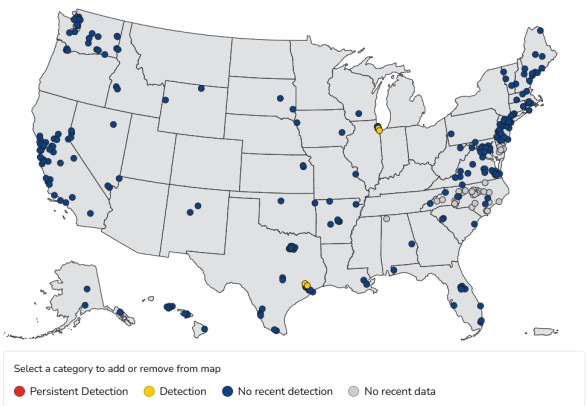


Source: Mpox activity - Wastewater monitoring dashboard — Canada.ca.

United States

The CDC's National Wastewater Surveillance System (NWSS) provides a public health framework to monitor infectious diseases through wastewater across the country. Wastewater testing for the mpox virus differs by location. Some locations test for specific clades, focusing on either clade II or clade I, while others use methods that detect both clades without distinguishing between them. Communities are voluntarily monitoring the presence of mpox in wastewater on a weekly basis. During the four-week period from August 24, 2025 - September 20, 2025, 8 out of 533 wastewater sites (2%) that tested for mpox reported the presence of the virus. Specifically, 4 sites indicated the detection of clade II. The remaining sites either reported no mpox detection in wastewater (480 sites) or had no recent data available (43 sites). The remaining sites either reported no mpox detection in wastewater (486 sites) or had no recent data available (44 sites).

Figure 29. Mpox virus detection in wastewater in the past 4 weeks, United States.



Source: U.S. Mpox Wastewater Data | National Wastewater Surveillance System | CDC.

Mpox news

24.10.2025

Local transmission of clade 1b mpox cases detected in EU/EEA, ECDC urges renewed vigilance

25.10.2025

Mpox - BBC News

27.10.2025

Latest update on Clade Ib mpox - GOV.UK

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EU Wastewater Observatory for Public Health

Mpox news

30.10.2025

Multi-country outbreak of mpox, External situation report #59 - 30 October 2025

31.10.2025

WHO reports rapid spread of Mpox with 17 deaths in Africa over recent weeks | Africanews

31.10.2025

Mpox: Latest News and Updates | South China Morning Post

Avian Flu (H5N1) surveillance

In addition to monitoring avian flu trends through wastewater surveillance data from national dashboards and National Health Authorities' websites, this section provides a comprehensive overview of recent Avian flu news.

Avian Flu in Wastewater: Since the beginning of 2025, avian influenza A(H5N1) has been increasingly detected in wastewater in various regions, especially in the United States. In Europe and worldwide, there has been a continued rise in outbreaks among wild birds and mammals, with new cases emerging even in remote areas, such as the sub-Antarctic islands.

Global Situation: Avian influenza A(H5N1) remains a significant global concern, with widespread outbreaks continuing among wild birds and poultry across multiple continents. According to the WHO, the global public health risk from H5N1 remains low, though occupational exposure risk is considered low to moderate depending on hygiene and biosecurity measures. Human infections are still primarily linked to direct contact with infected poultry or contaminated environments. There is no evidence of sustained human-to-human transmission.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

Avian Flu News

17.10.2025

Avian influenza

22.10.2025

Bird Flu Is Back - The New York Times

24.10.2025

Rapid early spread of bird flu in Europe raises fears of fresh crisis | Reuters

31.10.2025

Hungary reports bird flu outbreak on duck farm, WOAH says | Reuters

22.09.2025

Poland reports bird flu outbreak on farms in north, WOAH says | Reuters

26.10.2025

Outbreaks under monitoring: week 43 (week ending 26 October 2025) - GOV.UK

27.10.2025

Avian influenza overview June–September 2025 - - 2025 - EFSA Journal - Wiley Online Library

30.10.2025

Review: Human H5N1 avian flu cases can be asymptomatic, and the virus likely spreads among people | CIDRAP

Avian flu wastewater surveillance

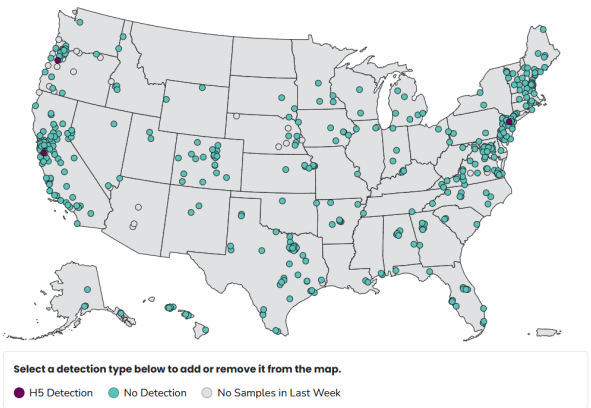
This section provides more recent information, gathered from dashboards and websites owned and operated by National Health Authorities, and listed on the EU Wastewater Observatory for Public Health.

NON-EU countries

United States. The CDC receives weekly data on the avian influenza A(H5) virus from a contract partner, an academic group, and several state health departments. Wastewater sampling and testing are conducted one or more times each week.

During week 43, avian influenza A(H5) results were reported from 432 sites. Of these, 3 sites (0.7%) across 3 states reported H5 detections, while 429 (99.3%) sites reported no detection.

Figure 30. Site-level data for avian influenza A(H5) viral RNA detections in wastewater.



Source: *Wastewater Data for Avian Influenza A(H5) | National Wastewater Surveillance System | CDC*.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

Poliovirus surveillance

This section provides a summary of latest detected news on poliovirus to keep informed about the most current developments.

Europe: Between September and December 2024, vaccine-derived poliovirus type 2 (cVDPV2) was detected in wastewater samples across multiple European countries, including Germany, Poland, Spain, Finland, and the United Kingdom. These detections are part of ongoing environmental surveillance efforts and no human polio cases have been reported in association with them. These detections primarily involve circulating vaccine-derived poliovirus type 2 (cVDPV2) genetically linked to a lineage first identified in Nigeria in 2020.

As of October 2025, no human cases of polio or poliovirus infections have been reported in association with these environmental detections in the EU/EEA, and the region remains free from the disease. However, the presence of poliovirus in wastewater highlights the need for continued vigilance and preventive measures, especially in under-vaccinated communities.

The European Centre for Disease Prevention and Control (ECDC) emphasises the importance of maintaining robust surveillance systems, including acute flaccid paralysis (AFP) monitoring and environmental surveillance, to rapidly detect and respond to any circulation of poliovirus.

Global update: Polio remains a global health concern, with wild poliovirus type 1 (WPV1) still endemic in Pakistan and Afghanistan, where 13 cases have been reported so far in 2025. Circulating vaccine-derived poliovirus type 2 (cVDPV2) continues to spread across 35 countries, with detections increasingly found in wastewater samples in regions including Europe, Africa, and Asia.

Poliovirus News

24.10.2025
Polio-free status retained, but European Region must increase vigilance

24.10.2025
'Let's finish the job' and end polio: WHO | UN News

24.10.2025
World Polio Day

29.10.2025
Polio this week – GPEI

Poliovirus Wastewater Surveillance

This section provides more recent information, gathered from dashboards and websites owned and operated by National Health Authorities, and listed on the EU Wastewater Observatory for Public Health.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

Poliovirus Wastewater Surveillance

EU countries

Belgium. Belgium has started monitoring poliovirus in January 2024. The surveillance program involves collecting wastewater samples every four weeks from the influent of 6 wastewater treatment plants. They do not plan to publish poliovirus data in a report.

Finland. Finland has been monitoring poliovirus in wastewater since the 1960s. Samples are collected biweekly in the Helsinki capital region and monthly in four other cities, covering approximately 30% of the Finnish population. At the moment, no data is being reported.

Ireland. Since February 2023, samples from the largest wastewater catchment area in Ireland have been tested every two weeks for poliovirus 2 (PV2). Currently, the focus is on PV2, based upon recent detections elsewhere of cVDPV2. Additional catchment areas and poliovirus types may be added over the coming months.

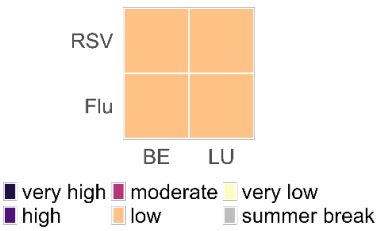
Slovenia. In 2024, Slovenia began monitoring for poliovirus in wastewater to detect any potential reintroduction of the virus into the country. This monitoring is conducted every three months at 16 WWTPs from all statistical regions, covering 34.2% of the population. The latest monitoring results, from September 2025, showed that all samples were negative for the virus.

Influenza, RSV and other viruses' surveillance

EU countries

This section provides information on RSV, influenza and other viruses wastewater surveillance derived from the latest wastewater surveillance data primarily obtained from EU national dashboards and websites owned and operated by National Health Authorities. Figure 30 illustrates the comparative viral loads of Influenza and RSV across various EU countries and includes only those countries for which indications on viral loads are provided.

Figure 31. Viral loads for Influenza and RSV across EU countries.



Source: EC-JRC.

Influenza

EU countries

Austria (regional data). Since October 2022, the influenza virus has been monitored weekly in the city of Vienna. he virus load is depicted in the graphics updated to week 43 and the data, including previous seasons, is available for download as an open governance dataset.

MONTHLY BULLETIN

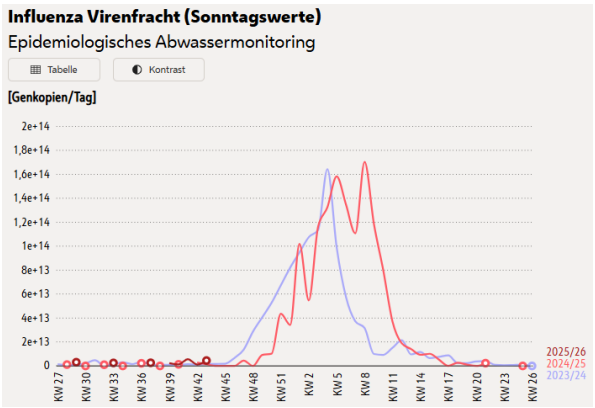
EU Wastewater Observatory for Public Health

Influenza

EU countries

Austria

Figure 32. Influenza virus load in Vienna.

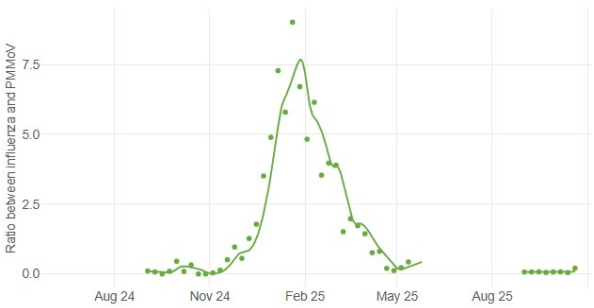


Source: Wastewater Monitoring - Weekly Data.

Belgium

Belgium has been monitoring the presence of influenza A and B since the beginning of this year. The surveillance program includes collecting wastewater samples once a week from the influent of 30 wastewater treatment plants. During week 43, the Influenza concentration was at a low level.

Figure 33. Ration between Influenza and PMMoV.

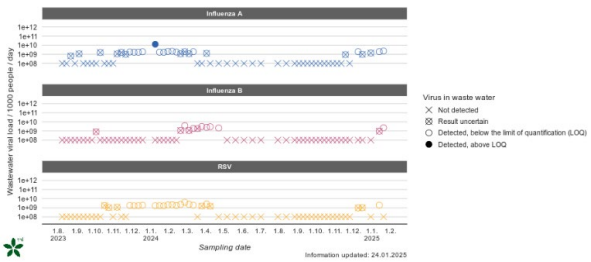


Source: Wastewater Sciensano.

Finland

The results for influenza A and influenza B, starting from July 31, 2023, are available at the municipal level for 11 locations. Quantitative results for 8 of these locations have been updated up to week 42 and can be downloaded in CSV format.

Figure 34. Influenza A and B viral loads in wastewater (top and middle panels), Helsinki.



Source: Influenza A, influenza B, and RSV in wastewater - THL

EU countries

Germany

Influenza A and Influenza B viral loads in wastewater are monitored at several wastewater treatment plants, and the number of reporting locations can fluctuate over time. For calendar week 41, data from 66 WWTPs were available. The average total viral load for Influenza A and B remained stable in the past 4 weeks. Additionally, monitoring for Influenza A and B in wastewater has been conducted in the German state of Bavaria since the beginning of 2024. Trends for Influenza are reported weekly on the local dashboard. During week 43, the viral load showed a decrease in 50% of the monitored locations, an increase in 40%, and remained stable in 10% of the locations.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

Influenza

EU countries

Germany

Figure 35. National influenza viral load in wastewater (gene copies per liter).



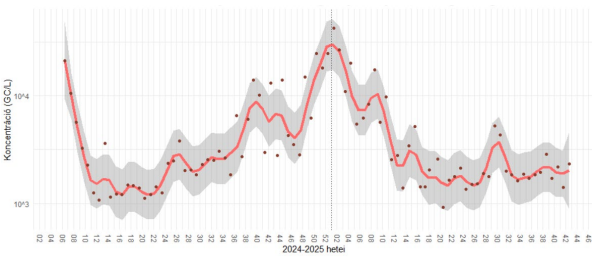
Source: *Infektionsradar*.

Hungary

The Influenza A virus has been detected in wastewater since January 2023. Starting from week 48 of 2024, the results for influenza A were supplemented with the amount of influenza B genetic material detected.

The sampling locations are the same as those used for SARS-CoV-2. In week 42, the national average concentration of influenza A and B genetic material has stagnated compared to the previous week.

Figure 36. National, population-weighted average of Influenza A copy number measured in wastewater.

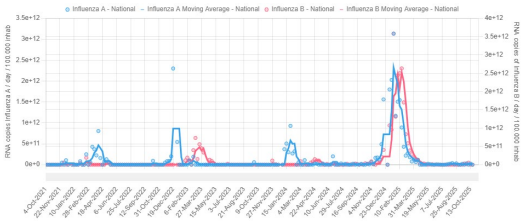


Source: *National Center for Public Health and Pharmacy - 2025.*

Luxembourg

The dashboard monitors the levels of Influenza A and B in wastewater from five cantons in Luxembourg, which represent about 67% of the population and include data from eight treatment plants. Results have been reported since October 2021, with updates every two weeks. As of week 43, the loads of Influenza A and B are lo

Figure 37. Influenza A/B national levels (RNA copies/day/100.00inhab).

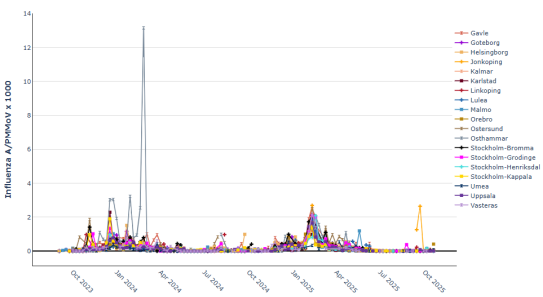


Source: *Influenza – MicrObs*.

Sweden

SLU-SEEC (the Swedish University of Agricultural Sciences) collects and analyses samples for influenza A and B viruses from 24 sites across Sweden, covering 43% of the population. The results are available up to week 43.

Figure 38. PMMoV normalised Influenza A content.



Source: *Amount of influenza virus in wastewater (SLU) | The Swedish Pathogens Portal*.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

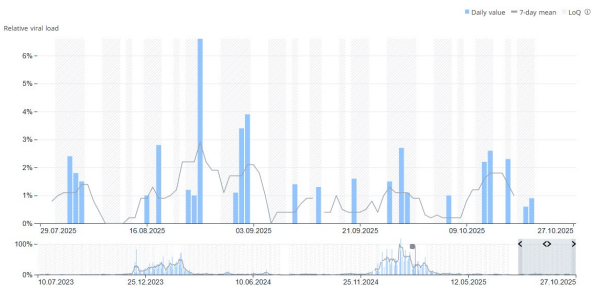
Influenza

Non-EU countries

Switzerland

Influenza A and B have been monitored through the national wastewater monitoring system since July 2023. Starting in January 2025, wastewater monitoring will be conducted at 10 participating WWTPs in Switzerland and one WWTP in the Principality of Liechtenstein. In week 43, the trends for Influenza A and B indicated a stagnation across all regions.

Figure 39. Development over time of influenza detections.



Source: IDD - Infectious Diseases Dashboard.

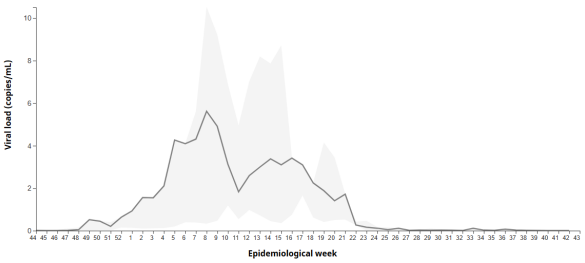
Canada

The dashboard provides information about the levels of Influenza A and B found in the wastewater from various communities across Canada.

In week 42, 19 sites are submitting sewage samples for analysis, which represents 29.2% of the Canadian population. The activity level of Influenza A was classified as low. Among the 19 tested sites, 5 sites showed a high activity level, 4 showed a moderate activity level, 2 showed a low activity level and 8 reported no detection.

For Influenza B, the activity level was classified as not detected because none of the 19 tested sites showed any detection.

Figure 40. National average of Flu A viral load (copies/ml) in wastewater.



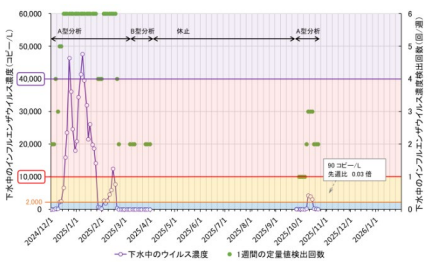
Source: Wastewater monitoring dashboard – Canada.ca.

Japan

In October 2024, Komatsu City began analysing and sharing information about the influenza A virus, providing updates 2-3 times a week. Influenza virus analysis is conducted from October to March.

During week 43, the levels of influenza A were low, with detection in 2 out of the 6 sewage samples, and the trend was decreasing.

Figure 41. Influenza A virus concentration in Komatsu City expressed in copies/liter.



Source: Influenza A Wastewater Data – Komatsu City

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

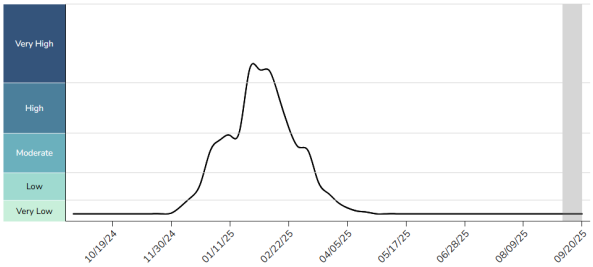
Influenza

United States

The CDC and public health partners are continuing to investigate whether the changing levels of influenza A virus in wastewater are related to seasonal influenza, the avian influenza A (H5) outbreak in animals, or both. Wastewater data cannot determine the specific source of influenza A viruses.

In week 38, the activity level of influenza A in wastewater across the country was very low.

Figure 42. National trends of wastewater viral activity levels of influenza A.



Source: Influenza A Wastewater Data – National Trends | NWSS | CDC.

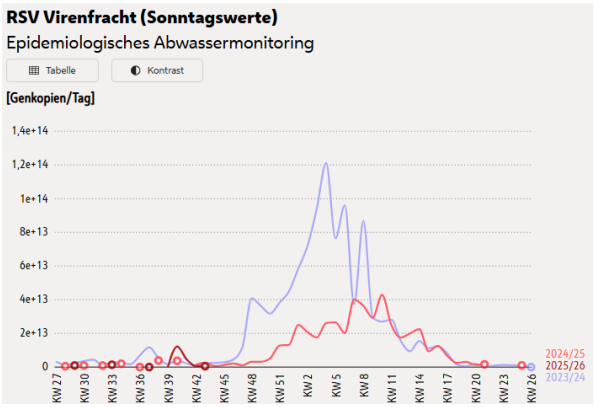
RSV

EU countries

Austria

(regional data). Since October 2022, the RSV has been monitored on a weekly basis in the city of Vienna. The virus load is depicted in the graphics updated to week 43, and the data, including previous seasons, is available for download as an open governance dataset.

Figure 43. Ratio between RSV and PMMoV.

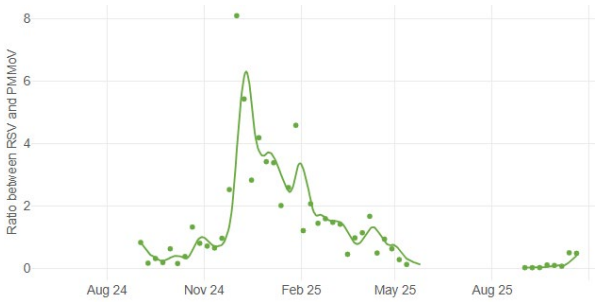


Source: Wastewater Monitoring - Weekly Data.

Belgium

Belgium started monitor RSV since the beginning of this year. The surveillance program involve the collection of wastewater samples once a week from the influent of 30 wastewater treatment plants. During week 42, the RSV concentration was at a low level.

Figure 44. Ratio between RSV and PMMoV



Source: wastewater sciensano

MONTHLY BULLETIN

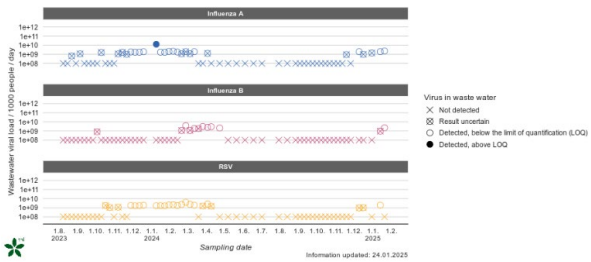
EU Wastewater Observatory for Public Health

RSV

Finland

The results RSV, starting from July 31, 2023, are available at the municipal level for 11 locations. Quantitative results for 8 of these locations have been updated up to week 37 and can be downloaded in CSV format.

Figure 45. RSV viral load in wastewater (bottom panel), Helsinki (Number of residents within the network: 860,000). (information updated: 24.01.2025).



Source: Influenza A, influenza B, and RSV in wastewater – THL.

EU countries

Germany

RSV viral loads in wastewater are monitored at several wastewater treatment plants, and the number of reporting locations can fluctuate over time. During reporting week 41, data from 42 WWTPs were available. The average viral load for RSV increased in the past 4 week. Since the beginning of 2024, wastewater monitoring for RSV has been conducted in Bavaria, Germany. Trends for RSV are reported weekly on the local dashboard. During week 43, the viral load showed a decrease in 30% of the monitored locations, an increase in 60%, and remained stable in 10% of the locations.

Figure 46. National RSV viral load in the wastewater (no. of gene copies per litre).

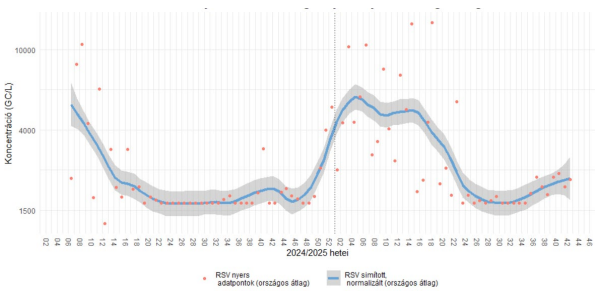


Source: Infection Radar.

Hungary

RSV virus has been detected in wastewater since January 2023. The sampling locations are the same as those for SARS-CoV-2. During week 42, the national average concentration of RSV showed a slight increase compared to the previous week and was above the detection limit at three sampling locations.

Figure 47. National, population-weighted average of RSV copy number measured in wastewater.



Source: National Center for Public Health and Pharmacy - 2025. Week 42 wastewater test results.

MONTHLY BULLETIN

EU Wastewater Observatory for Public Health

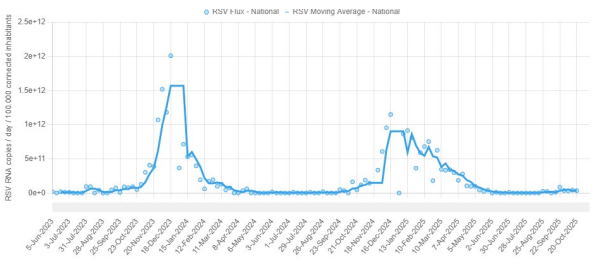
RSV

EU countries

Luxembourg

The dashboard monitors the levels of RSV in wastewater from five cantons in Luxembourg, which represent about 67% of the population and include data from eight treatment plants. Results have been reported since June 2023, with updates every two weeks. As of week 43, RSV levels are low.

Figure 48. RSV national levels (RNA copies/day/100.00inhab).

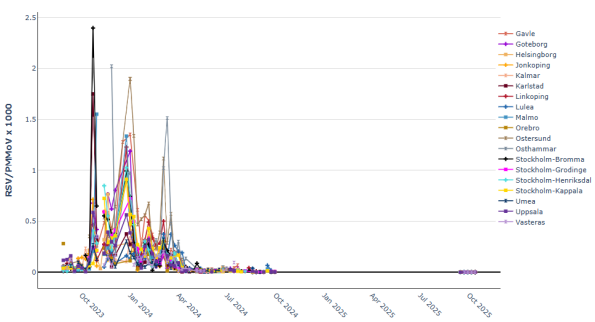


Source: RSV – MicrObs.

Sweden

SLU-SEEC (the Swedish University of Agricultural Sciences) collects and analyses samples for RSV from different areas of Sweden. The results are available up to week 42 for 19 cities.

Figure 49. RSV concentration in copies/liter.



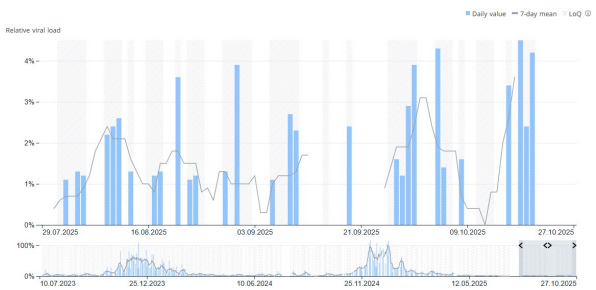
Source: Amount of Respiratory Syncytial Virus (RSV) in wastewater (SEEC-SLU) | The Swedish Pathogens Portal.

Switzerland

RSV has been monitored through the national wastewater monitoring system since July 2023. Starting in January 2025, wastewater monitoring takes place at 10 participating WWTPs in Switzerland and one WWTP in the Principality of Liechtenstein.

As of week 43, the RSV trends have been stagnant across all 11 regions.

Figure 50. RSV relative viral load for a selected region (Lugano).



Source: IDD - Infectious Diseases Dashboard.

Canada

The dashboard presents data on the levels of RSV detected in the wastewater of various communities across Canada. Currently, 19 sites are submitting sewage samples for analysis, representing 29.2% of the Canadian population. In week 42, the RSV activity level was classified as moderate. Of those sites, 6 showed a high activity level, 6 showed a moderate activity level, 1 showed a low activity level and 6 had no detection.

MONTHLY BULLETIN

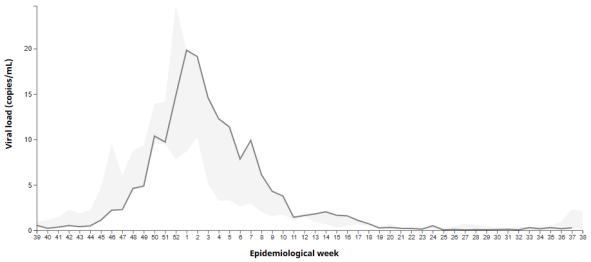
EU Wastewater Observatory for Public Health

RSV

Non-EU countries

Canada

Figure 51. National average of RSV viral load (copies/ml) in wastewater.



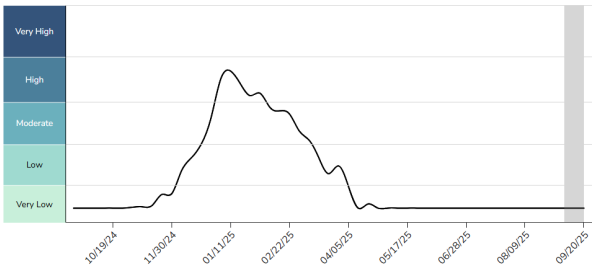
Source: Wastewater monitoring dashboard – Canada.ca.

United States

CDC's National Wastewater Surveillance System (NWSS) provides the public health infrastructure to monitor RSV through wastewater across the country.

During week 38, the level of viral activity for RSV in the wastewater was very low on a national level.

Figure 52. National trends of wastewater viral activity levels of RSV.



Source: RSV Wastewater Data – National Trends | NWSS | CDC.

Enteric Viruses

EU countries

Luxembourg

Norovirus and Enterovirus have been monitored in Luxembourg from March 2020 to July 2024. Wastewater levels were reported from two cantons, covering approximately 52% of the population through four treatment plants. Latest sampling date: 15.07.2024.

Sweden

Norovirus and Enterovirus have been monitored in Luxembourg from March 2020 to July 2024. Wastewater levels were reported from two cantons, covering approximately 52% of the population through four treatment plants. Latest sampling date: 15.07.2024.

EDITORIAL OFFICE
Joint Research Centre, Via E. Fermi, Ispra, Varese (VA)
email: JRC-EU4S-DEEP@ec.europa.eu
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Blue Mirror

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